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Post-Crisis Valuation of Derivatives

Doctoral Thesis

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Declaration

I declare that I wrote this doctoral thesis independently and that all external sources and literature are acknowledged in the text.

Prague, February 15, 2016 Jaroslav Baran

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Abstract

In this study we analyse relationship between classical approach to valuation of linear interest rate derivatives and post-crisis approach when the valuation better reflects credit and liquidity risk and economic costs of the transaction on top of the risk-free rate. We discuss the method of collateralization to diminish counterparty credit risk, its impact on derivatives pricing, and how overnight indexed swap (OIS) rates became market standard for discounting future derivatives' cash flows. We show that using one yield curve to both estimating the forward rates and discounting the expected future cash flows is no longer possible in arbitrage free market. We review in detail three fundamental interest rate derivatives (interest rate swap, basis swap and cross-currency swap) and we derive discount factors used for calculating the present value of expected future cash flows that are consistent with market quotes. We also investigate drivers behind basis spreads, in particular, credit and liquidity risk, and supply and demand forces, and show how they impact valuation of derivatives. We analyse Czech swap rates and propose an estimation of CZK OIS curve and approximate discount rates in case of cross-currency swaps. Finally, we discuss inflation markets and consistent valuation of inflation swaps.

Keywords: swap spreads, overnight indexed swap, basis swap, cross-currency swap, basis spread, discount factor, collateral

JEL Classification: D53, G01

Abstrakt

V této práci se zabýváme vztahy mezi klasickým přístupem k oceňování lineárních úrokových derivátů a postkrizovým přístupem, který kromě diskontování bezrizikovou křivkou rovněž zohledňuje kreditní a likviditní riziko a ekonomické náklady transakce. Pojednáme o tom jak lze pomocí kolateralizace snížit kreditní riziko protistrany, jaký vliv má kolateralizace na oceňování derivátů a jak se OIS sazby stali tržním standardem pro diskontování budoucích peněžních toků derivátových transakcí. Ukážeme, že používání jedné křivky pro odhad forwardových sazeb a zároveň pro diskontování očekávaných peněžních toků již není možné na bezarbitrážním trhu. Podrobně rozebereme tři základní úrokové deriváty (úrokový swap, bazický swap a meziměnový swap) a odvodíme diskontní faktory používané k výpočtu současné hodnoty očekávaných budoucích plateb, které jsou konzistentní s tržními kotacemi. Dále zkoumáme bazické přirážky z pohledu kreditního a likviditního rizika a nabídky a poptávky a ukážeme, jak tyto přirážky ovlivňují oceňování derivátů. Analyzujeme swapové sazby na českém trhu a ukážeme, jak lze odhadnout korunové OIS sazby a aproximovat diskontní sazby pro ocenění meziměnových swapů. Nakonec prodiskutujeme inflační trhy a oceňování iniflačních swapů.

Klíčová slova: swapové přirážky, overnight indexed swap, bazický swap, meziměnový swap, bazické přirážky, diskontní faktor, kolaterál

JEL Klasifikace: D53, G01

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1. Introduction

Risk-free rates are the essential input in calculation of present value of expected future cash flows and in general, for pricing of financial instruments. In practice, to calculate a risk-free yield curve, proxies such as government bond yields and short-term interbank deposit interest rates (see Málek et. al (2007)) have been used. For valuation of derivative instruments using quoted forward rates (FRA, Forward Rate Agreement) and swap rates (IRS, Interest Rate Swap) have been a standard approach in the past.

The basis of these instruments is the exchange of fixed for floating rate cash flows. The floating rate is usually the 3-month or 6-month IBOR (interbank offered rate) and reflects the unsecured interbank funding rate for this maturity. IBOR rates therefore also include the average risk premium of the given interbank market or respective reference banks. This risk premium is also a part of the quoted swap rates and is thus reflected in the swap curve. Under normal market conditions, the risk embedded in short-term interbank loans is considered very low and the swap curve has been used as a proxy of a risk free curve; however, this no longer holds, especially during market stress.

The increase in interbank risk has become a major issue during financial crisis with a direct impact on banks' liquidity management. We perceive interbank risk mainly as credit and liquidity risk that are integral components of unsecured interbank loans. IBOR market reference rates which reflect the interbank risk are one of the main channels of monetary transmission mechanism and serve as a benchmark for wide range of financial products, i.e. financial derivatives, mortgages, loans, or savings accounts. Since the breakout of recent financial crisis, it has been shown that these rates are not adequate proxies for risk-free rates, see for example Bianchetti (2008) or Bianchetti and Carlicchi (2012).

The risk premium in quoted IBOR rates is incorporated into quoted FRA and IRS rates, and is therefore reflected in the yield curve derived from these instruments. According to Collin-Dufresne and Solnik (2001), the swap rate is characterized as fixed rate equivalent of a continuous short-term borrowing by banks of high credit rating (the average rating of IBOR panel banks). If a rating of a certain bank deteriorates, the bank will be replaced by another highly rated bank. During normal market conditions, the risk premium in interbank rates has been considered as negligible and IBOR swap rates were therefore used as a proxy to construct a risk-free yield curve. At the same time, IBOR swap rates were a good proxy for banks' funding costs. Nevertheless, IBOR swap rates are not risk-free anymore nor they approximate banks' funding costs any longer.

We may look at interbank rates and compare them with yields of government t-bills that are regarded as the least risky financial instrument. For USD rates, such indicator is called TED spread and is defined as the difference between 3-months USD LIBOR rate and 3-months US T-bills yield; it shows the risk premium of commercial banks to US government. Long-term historical average of this indicator until 2007 had been 30 basis points, however, during crisis it jumped to 450 bps (Figure 1, LHS). The same happened in other countries and markets.



Figure 1: Historical development of 3M USD Libor, 3M USD T-Bills and their difference (TED spread, LHS), and 3M USD Libor, 3M USD OIS swap rates and their difference (Libor-OIS spread, RHS) since 2005. *Source: Bloomberg*

It has become apparent that interbank rates are no longer valid proxies for risk-free rates due to both their significant risk premium and their volatility. In fact, it has been argued that OIS (overnight indexed swap) rates are a better proxy for risk-free rates. OIS swaps are based on the exchange of fixed rate against floating overnight (O/N) reference rate of respective interbank market. Although both IBOR and OIS swap rates are based on interbank risk, overnight deposits bear lower risk than term deposits; therefore the risk in OIS rates is lower than the risk in IBOR rates (Figure 1, RHS).

In fact, the floating rate in OIS swap is based on overnight unsecured interbank rates (it equals geometric average of overnight rates). OIS swap rates therefore reflect market expectations of average overnight rates of reference banks in the respective interbank market. If a credit quality of a reference bank worsen and the bank's credit rating is downgraded, such bank may drop from panel banks contributing to the overnight index calculation. Therefore the credit risk of the OIS swap rate is limited to the risk of default of a bank with similar credit quality as of those from the panel banks in the horizon of one day. Annualized probability of such default is negligible and in any case significantly lower than the annualized probability of bank's default in a three or six month's horizon (3-month or 6-month IBOR rates).

When we look at cross-currency derivatives, we notice another interesting phenomenon; the existence of cross-currency basis spread to a floating rate of one currency vs. the other. This spread can be explained by different liquidity and credit risk of interbank market in one currency vs. the other, and by demand and supply imbalances. We discuss the existence of credit and liquidity risk premiums on top of risk-free rates in Baran and Witzany (2014). The existence of basis swap spreads itself leads to discrepancies in their interpretation. According to Chang and Schlogl (2012), basis swap spreads are inconsistent with a classical arbitrage argument between spot and forward market. On the other hand, Bianchetti and Carlicchi (2012) note that it is consistent with arbitrage-free market with the consequence that the valuation of such derivatives needs multiple curve input for estimating forward rates and discounting future cash flows. In fact, when we change the discount curve, we change the market value of the derivative. The concept of using one curve to both estimate the forward rates and discount future cash flows thus needs to be reassessed.

And this is the aim of our study. In this work, we analyse relationship between classical approach to valuation of linear interest rate derivatives and post-crisis approach that requires further assumptions to reflect the risk of the transaction, for example counterparty credit risk. We discuss the method of collateralization to diminish counterparty credit risk and its impact on derivatives' pricing. We also analyse credit and liquidity risks, and supply and demand forces and their impact on valuation of derivatives, in particular, interest rate swaps, basis swaps and cross-currency swaps. We show that using one yield curve for both estimating forward rates and discounting expected future cash flows is no longer possible in arbitrage free market. We present valuations of three fundamental interest rate derivatives (interest rate, basis and cross-currency swaps) step by step and we derive discount factors used to calculate the present value of expected future cash flows that are consistent with market quotes.

This work is organized into six chapters. Each chapter starts with an introduction and a review of relevant literature. The second chapter discusses recent changes and trends in derivatives' valuation when different adjustments are needed to reflect risks and costs of a transaction, for example, counterparty credit risk, funding costs, or capital charges. We discuss valuation methodologies when future cash flows are discounted with risk free rates and such valuation is then adjusted to take into account all economic costs of the transaction. The topic of valuation adjustment is still evolving and as we discuss later, not without controversy.

The third chapter is the core chapter of the thesis. In this chapter we discuss derivatives' discounting in detail. We explain how OIS rates became a market standard for discounting future cash flows due to being both a good proxy to risk-free rates and due to the fact that they are linked to interest paid on collateral for collateralized transactions. OIS discounting takes the centre stage but we also discuss other discount curves. The aim of this chapter is to derive

discount factors and forward rates that are consistent with market prices. In this respect, we also show how basis swap spreads impact the valuation. We focus in detail on valuation of interest rate swaps, basis swaps and cross-currency swaps. We show that we can no longer work with one curve but use multiple curves to estimate forward rates and to discount future cash flows. Concluding section of this chapter deals with approximations of discount curves implied from cross-currency market quotes and the discount curve of the base currency. Apart from technical derivations, we also further investigate the relationship between discounting and forwarding curves of both legs of cross-currency swap and discuss potential implications and applications.

Fourth chapter is more practical and targets the Czech derivative market. We discuss how CZK OIS curve could be estimated even if not all OIS quotes are available. We also analyse EUR/CZK cross-currency basis swap spreads by decomposing them into individual components and investigate if they are driven mainly by different credit and liquidity risk between EUR and CZK interbank rates or by relative demand of one currency versus the other. We also show how we can use such breakdown to approximate implied discount rates of the second leg of the cross-currency swap.

Fifth chapter analyses cross-currency basis swap spreads from a different angle. We discuss credit and liquidity risk and supply and demand pressure of one currency versus another one. We then build a multiple regression model and explain drivers behind EUR/USD basis swap spreads. We construct boundaries within which there should not be an arbitrage opportunity, however, by testing these boundaries we discover that supply and demand imbalances may push basis spreads outside these boundaries creating arbitrage opportunities for market participants able to raise unsecured funding in one currency and swapping it into another currency. We illustrate this on the example of EUR/USD and EUR/CZK FX swap markets.

The last, sixth, chapter differs slightly by focusing on inflation market and inflation derivatives. One of the reasons to include this topic is that inflation markets do not enjoy such spotlight as other market although they are quite sizeable and new valuation methods apply to them as well as to other derivatives. That is why the aim of this chapter remains the same; to present a consistent valuation approach to zero-coupon inflation swaps reflecting the risk of its future cash-flows. However, we also provide a general overview on functioning of inflation markets, and we discuss inflation measurement, real yields and nominal yields, and the construction of inflation curve from zero coupon inflation swaps taking into account inflation seasonality.

Some results of the thesis were published in journals or presented at conferences, in particular, chapters 3 and 4 were published together with J. Witzany in journal *Politická ekonomie: teorie modelování, aplikace* (2014. sv. 62, č. 1, s. 67-99) and presented at the *6th CSDA International Conference on Computational and Financial Econometrics (CFE 2012)* in 2012 in Oviedo, Spain.

2. Valuation adjustments for derivatives

IBOR rates were in the past a good proxy for valuing derivatives for being close enough to risk-free rates and also reflecting interbank risk. Recent years have changed the valuation of derivatives as other risks have emerged from being negligible to being impactful to pricing. The presence of these risks brought several adjustments to risk-neutral valuation of derivatives.

Derivatives are transacted between counterparties that are not risk-free. They bear, for example, counterparty risk as either party may default during the life of the trade. Moreover, derivative positions need to be funded and such funding costs are often reflected in the pricing. On top of that, banks do not shy away from passing on capital charges to their clients. Derivatives are thus being increasingly priced including all their economic costs. One way to reduce these costs is to increase collateralization of trades. In fact, proper calculation of market values of fully collateralized derivatives is the main focus of this thesis but without this chapter, the thesis would not be complete.

In this chapter we discuss recently introduced valuation adjustments made to the value of derivatives to adjust for their economic costs, in particular credit risk, funding costs, or capital costs. Namely, we will discuss credit valuation adjustment (CVA), debit valuation adjustment (DVA), funding valuation adjustment (FVA), capital valuation adjustment (KVA), and initial margin valuation adjustment (MVA). All valuation adjustments are covered by a collective term XVA introduced by Carver (2013). As we discuss below, most of these adjustments are linked to either uncollateralized derivative transaction or the uncollateralized part of the derivative transaction.

From an organizational point of view, XVA valuations function sometimes stays on the trading desk and sometimes with the treasury department but the current trend seems to be that banks set up centralized XVA desks that manage all valuation adjustments to derivatives in addition to their risk-free fair value. XVA desk calculates counterparty credit risk, capital costs, and other charges to each derivative trade to capture its real costs. It then allocates these costs to different desks. Despite their different meaning, the XVA valuation adjustments follow similar valuation principles. Recently published book by Gregory (2015) discusses latest market practice on XVA valuation adjustments and tackles most of actual issues from the practitioner's point of view.

2.1. Credit valuation adjustment (CVA)

CVA quantifies the counterparty credit risk; it is the difference between the market value of a derivative contract with counterparty with specific credit risk and the market value of the same derivative with a risk-free counterparty. With unilateral CVA calculation we only take into account credit risk of the counterparty and assume we are risk-free. CVA thus serves both to calculate the fair value of the transaction (risk-free valuation plus the counterparty credit risk) and by building a CVA buffer it also serves as a form of insurance against the counterparty credit risk and its volatility.

CVA price depends on both credit spread of the counterparty as well as other market risk factors that may impact the value of the derivative. A common approach to calculate credit adjustments is to simulate future value of the derivative and calculate the expected exposure over its life. The following text is based on Zhu and Pykhtin (2007) to illustrate the calculation of the CVA.

Taking into account counterparty credit risk means calculating the discounted expected loss if counterparty defaults. The discounted loss in case of counterparty's default before the maturity date and after recovering percentage *R* can be written as

$$L = (1 - R)DF_\tau \xi_\tau \tag{2.1}$$

where τ is the time of counterparty's default, ξ_τ is the counterparty exposure at time τ, and DF_τ is the risk-free discount factor to calculate present value of the loss. The CVA is then the risk-neutral expectation of the discounted loss from (2.1) conditional on counterparty's default occurring at time *t*, in other words, an integral over the whole life of the transaction of the discounted expected exposure conditional on counterparty's default and multiplied by the probability of default at time *t* and by one minus recovery rate *R*

$$CVA = (1 - R) \int_0^T E^Q(DF_t \xi_t | \tau = t) d\Pi_{0,t} \tag{2.2}$$

where $E^Q(DF_t \xi_t | \tau = t)$ is the expected discounted exposure conditional on counterparty's default at time t, $\Pi_{0,t}$ is the risk neutral probability of counterparty's default between time 0 and *t*, and *T* is the final maturity date of the transaction. A liquid CDS market can be used to extract these probabilities from CDS spreads of the counterparty at different maturity dates. If CDS market does not exist, a rating proxy can be used. The formula simplifies if there is no correlation between exposure and counterparty's credit risk

$$CVA = (1 - R) \int_0^T DF_t \xi_t d\Pi_{0,t} \tag{2.3}$$

The expected exposure is numerically simulated for each future date, and (2.3) thus in practice becomes the sum

$$CVA \approx (1-R)\sum_{k=1}^{n} E\xi_{t_k}\Pi_{t_{k-1},t_k}. \tag{2.4}$$

where Π_{t_{k-1},t_k} is the probability of default between dates t_{k-1} and t_k, and $E\xi_{t_k}$ is the expected exposure at date t_k.

In the above formulas we have used the term risk-neutral probability. For the purpose of valuation of derivatives, it is often assumed that the present value of a derivative equals the future expected payoff that is discounted at the risk-free rate and there is no arbitrage. Therefore it is assumed that neither counterparty will default.

2.1.1. Modelling counterparty exposure

A credit exposure to counterparty in a single derivative is either zero in case of negative market value or the positive market value of the derivative. Let ξ_t represent this exposure at time *t*

$$\xi_t = max[V_t, 0]$$

where V_t is the market value of a single derivative at time t. For a portfolio including netting, the total exposure is then the net portfolio value or zero, if the net portfolio value is negative

$$\xi_t = max\left[\sum_{i=1}^{n} V_{i,t}\,, 0\right]$$

where $V_{i,t}$ is the market value of derivative *i* at time t. At time t we only know the current exposure. To calculate counterparty credit risk and the required regulatory capital we need to model the potential future exposure.

Future exposure is uncertain and is usually modelled via Monte Carlo simulation of the evolution of market risk factors (e.g. interest rates, FX rates) at each future date t_k, because the default can occur at any date in the future. If exposure is independent of counterparty's credit risk and there are no collateral agreements, then based on the generated values of market risk factors we calculate the market value of each derivative V_{i,t_k} at each future date t_k, the portfolio value $V_{t_k} = \sum_{i=1}^{n} V_{i,t_k}$, and exposure to counterparty $\xi_{t_k} = max\left[V_{t_k}, 0\right]$. Discounted expected exposure is then calculated as the average of a large number of such scenarios at each future date t_k. For *N* scenarios this leads to expected exposure at date t_k

$$E\xi_{t_k} = \frac{1}{N}\sum_{j=1}^{N} DF_{t_k}^{j}\xi_{t_k}^{j} \tag{2.5}$$

and CVA is then calculated as in (2.4)

$$CVA = (1-R)\sum_{k=1}^{n} E\xi_{t_k}\left(\Pi_{t_k} - \Pi_{t_{k-1}}\right)$$

$$= (1-R)\sum_{k=1}^{n}\frac{1}{N}\sum_{j=1}^{N} DF_{t_k}^{j}\xi_{t_k}^{j}\left(\Pi_{t_k} - \Pi_{t_{k-1}}\right) \quad (2.6)$$

where Π_{t_k} is the cumulative probability of default up to date t_k.

If the derivative is subject to collateral agreement, the value of collateral C_{t_k} received at time t_k will decrease the exposure of each derivative $\xi_{t_k} = max\left[V_{t_k} - C_{t_k}, 0\right]$.

For more detailed discussion on CVA calculations including collateral agreements that depend on credit quality of the counterparty we refer to Pykhtin and Rosen (2010). For a discussion on exposure dependency on counterparty's credit quality (wrong-way risk) we refer to Černý and Witzany (2013) and Černý and Witzany (2015).

2.1.2. Regulatory CVA

Basel III framework (Basel (2011)) argues that the volatility of the counterparty credit quality leads to greater losses than the actual counterparty defaults and requires banks to set aside CVA capital provision to cover potential mark-to-market losses resulting from lower credit worthiness of the counterparty.

European banks implement Basel III regulatory standards via EU directive and regulation, namely the Capital Requirements Directive (CRD IV), and the Capital Requirements Regulation (CRR)[1]. Under the new capital requirements, banks need to calculate CVA capital charge for all OTC derivatives to build a buffer in case the counterparty credit worthiness deteriorates. Derivatives cleared via central counterparties are excluded from CVA charge.

Banks may be granted permission to calculate CVA charge using their internal models which simulate changes in the credit spreads of counterparties. If banks are approved to use internal models, then they must use the *Advanced Method* to calculate CVA charge. Under advanced approach, banks simulate counterparty credit spreads over a ten business day horizon, apply a CVA formula to each counterparty taking into account CVA hedges and calculate 99% Value-at-risk (VaR) of the distribution of CVA losses. CVA is calculated according to the following formula

$$CVA = LGD_{MKT}\sum_{i=1}^{T} max\left\{0, e^{-\frac{s_{i-1}t_{i-1}}{LGD_{MKT}}} - e^{-\frac{s_i t_i}{LGD_{MKT}}}\right\}\frac{E\xi_{i-1}DF_{i-1} + E\xi_i DF_i}{2}$$

[1] CRD IV entered into force on 17 July, 2013, and came into effect on 1 January, 2014.

where t_i is the time of *i*-th revaluation, s_i is the credit spread of counterparty at tenor t_i, LGD_{MKT} is market implied or a proxy of loss given default of the counterparty, and $E\xi_i$ is the expected exposure to the counterparty at time t_i discounted with risk-free discount factor DF_i. The factor in the middle represents marginal probability of default between times t_{i-1} and t_i.

CVA risk capital is then calculated as the sum of non-stressed and stressed (with shifted credit spreads) CVA VaR multiplied by 3

$$CVA_{advanced} = 3(VaR_{non-stressed} + VaR_{stressed})$$

If a bank does not use an internal model, then it is required to use the *Standardized Method*[2] according to the following formula that is based on one-year VaR approach

$$CVA_{standard} = 2.33\sqrt{h}\sqrt{\left(\sum_{i=1}^{n} 0.5w_i\left(M_i\, EAD_i^{total} - M_i^{hedge} B_i\right) - \sum_{ind=1}^{n} 0.5w_{ind} M_{ind} B_{ind}\right)^2 + \sum_{i=1}^{n} 0.75w_i^2\left(M_i\, EAD_i^{total} - M_i^{hedge} B_i\right)^2}$$

where h is the one-year risk horizon, $\mathrm{w_i}$ and $\mathrm{w_{ind}}$ are weights of counterparty i or index ind based on their credit risk, EAD_i^{total} is the total counterparty i credit risk exposure, M_i, M_i^{hedge}, and M_{ind} are effective maturities of transactions with counterparty i , of the hedge instrument, and of the index hedge, and B_i and B_{ind} are notionals of purchased single name and index CDS contracts used to hedge the CVA risk. These are permitted eligible hedges for the purpose of calculating the regulatory capital charge.

Douglas and Pugachevsky (2012) show that the standardized formula can be interpreted as a one-year 99% CVA VaR with normal distribution with correlations between assets equal to 25% and correlations with the index being 50%.

Both methods are one-sided (unilateral) CVA methods. This means that the framework does not recognize the probability of bank's own default and assumes it is risk-free (the DVA component).

2.2. Debit valuation adjustment (DVA)

Debit valuation adjustment adjusts the price of the derivative for the risk of one's own default. Despite some controversy, it is now a common practice to take into account both CVA and DVA when pricing a derivative, that is, to adjust its price for both the risk that the counterparty may

[2] Article 384 of the CRR (http://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32013R0575)

default and the risk of one's own default. The controversy comes from the accounting recognition of DVA gains if one's own credit quality deteriorates (and DVA increases).

While in the case of CVA we calculate the positive expected exposure in case the counterparty defaults, in the case of DVA we calculate the negative expected exposure in case of our own default. The probabilities of default in case of CVA or DVA are likely to be different, too.

Assuming no correlation between exposure and one's own credit risk, we can express DVA as in (2.3)

$$DVA = (1 - R) \int_0^T DF_t \xi_t^- d\Pi_{0,t} \tag{2.7}$$

where ξ_t^- is the negative exposure $\xi_t^- = max[-V_t, 0]$, or, the exposure from the view of the counterparty (instead of positive exposure profile we take the negative exposure profile), and *R* is the recovery rate in case of one's own default.

Bilateral CVA

Derivatives have bilateral credit risk, thus bilateral CVA. Bilateral CVA can be seen as the difference between CVA and DVA, which need to be modelled jointly. Value of a derivative after adjusting for credit risk of both parties thus becomes

$$V = V_{risk-free} - CVA + DVA$$

$$V = V_{risk-free} - \left(1 - R_{ctpty}\right) \int_0^T DF_t \xi_t^+ d\Pi_{0,t}^{ctpty} + (1 - R_{own}) \int_0^T DF_t \xi_t^- d\Pi_{0,t}^{own} \tag{2.8}$$

where *ctpty* denotes parameters linked to counterparty's default and *own* denotes the case of one's own default. Bilateral CVA thus nets the credit risk of two opposite counterparties.

2.3. Funding valuation adjustment (FVA)

We have discussed above that banks no longer borrow at interbank rates but at very individual rates that reflect their riskiness. These individual funding costs are often taken into account when calculating the price of the uncollateralized part of a derivative. If a fully collateralized trade is hedged with uncollateralized one then posting collateral in the first trade is not offset by receiving collateral in the second trade and the collateral is funded at bank's individual rates. If these rates are higher than the rate on the posted collateral, the bank bears extra funding costs and attributes these costs on top of the CVA/DVA of the derivative. The reasoning is that funding costs are different and usually higher than bank's credit risk as they also include funding liquidity premium (the actual interest rate bank is paying on its unsecured borrowings is higher than the CDS premium).

Assuming the collateral is remunerated at a risk-free rate, the funding valuation adjustment is the incremental spread one pays on top of the risk-free rate and its own credit spread (from the DVA) to fully reflect one's borrowing costs

$$FVA = \int_0^T DF_t \xi_t^- s_t \, dt \tag{2.9}$$

where s_t is the funding spread on top of one's own credit spread and other parameters are the same as for the DVA. FVA thus takes into account one's own cost of funding on top of the risk-free rate and DVA (not to double count the benefit of the DVA). FVA modifies the equation (2.8) as follows

$$V = V_{risk-free} - CVA + DVA - FVA \tag{2.10}$$

On the other hand, if the bank receives collateral from the hedge counterparty but does not post this collateral in the other uncollateralized trade, the dealer should be able to lend this collateral at and an individual lending rate that creates a benefit for the FVA calculation. FVA in equation (2.9) is often referred to as Funding Cost Adjustment (FCA), while reversing sides and earning extra return on received collateral is reflected in the equation (2.10) with the positive sign and referred to as Funding Benefit Adjustment (FBA, calculated as in (2.9) but with negative expected exposure). For technical details on FVA calculation, we refer to Pallavicini et al. (2011) who suggest least square Monte Carlo techniques to price a derivative adjusted for CVA, DVA, and margining and funding costs.

FVA is thus different from the credit risk component that is incorporated in CVA/DVA calculations. There is an apparent overlap between bank's own credit risk (DVA) and funding costs and also it is not straightforward to determine precise funding costs. One way to calculate FVA and not to double count the DVA is to use the spread between bank's unsecured bonds' risk premium and its CDS premium but this approximation has its drawbacks, for example, it can be negative. Using one's own funding costs in (2.9) has not been the only development in FVA, in fact, a lot of ambiguity to its calculation persists. Some argue that it is not one's own funding costs that should be reflected in this case, but funding cost of the counterparty (on top of their credit risk, which may be even more difficult to estimate) or some kind of average market funding costs (to reflect a market view).

Including funding costs and benefits into (2.10) brings more controversy than DVA. For example, Hull and White (2012c) argue that for valuation or pricing of a derivative FVA should be ignored as neither the hedging nor dealer's funding (liquidity) costs should impact the value of a derivative. Derivative value should always reflect its risk, not how it is funded. Practitioners' views differ and often question the basic assumptions such as borrowing at risk-free rates as markets are incomplete. For example, Ruiz (2015) argues that the counterparty credit risk (or

CDS spread if it is liquid) can be different from bank's cost of borrowing and cost of collateral. Burgard and Kjaer (2013) show how different funding strategies and spreads may create different values of derivative positions. Interestingly, Hull and White (2014) admit that one may defend adjusting the valuation for the liquidity component of bank's credit spread but continue to argue that FVA may result in different estimates of market value by different counterparties with different funding costs. Authors continue to advocate a single price for any derivative that can be reached by properly calibrated and incorporated CVA and DVA, but not FVA, and further provide an example of arbitrage if FVA valuations are taken into account. Nevertheless, there seems to be a trend that banks include FVA both in pricing new trades and valuation for accounting purposes. This may be linked to a fact that derivatives' dealers are being charged bank's funding costs by their treasury or XVA desk and not the risk-free rate. This is why Hull and White (2014) propose that treasury should lend funds to trading desks at the risk-free rate. Nevertheless, if banks include FVA into their pricing, then those banks with higher credit quality and lower funding costs may be more competitive to price uncollateralized derivatives that require funding. Banks currently do not have to recognize FVA in terms of fair value accounting nor there are any regulatory requirements for FVA and the FVA discussions continue but yet without an agreement on how to exactly determine funding costs.

2.4. Other valuation adjustments

Other add-ons are being increasingly included in derivative's value alongside CVA, DVA or FVA. They are calculated in a similar manner. We discuss two of them that we consider the most important, namely, Capital valuation adjustment and Margin valuation adjustment.

2.4.1. Capital valuation adjustment (KVA)

Regulatory changes for banks' capital requirements have recently driven banks to include these capital requirements into the valuation of derivatives. KVA adjustment calculates the cost of such regulatory capital that would act as a buffer for unexpected losses and requires estimating capital needs of a derivative over its life. In practice, usually treasury charges capital costs to the dealer's desk. This term was introduced and formalized by Green et al. (2014) to whom we also refer for details on technical computational details. KVA usually comes as a cost charged to a counterparty and thus modifies the formula (2.10) as follows

$$V = V_{risk-free} - CVA + DVA - FVA - KVA \tag{2.11}$$

A complete price thus includes its risk-free valuation (or fully collateralized valuation), counterparty credit risk, one's own credit risk, unsecured funding costs and capital charges.

Sherif and Chambers (2015) conducted a survey on KVA in July 2015 showing interesting results that most of banks now price KVA at least partially into trades and banks identify KVA to be the biggest from the XVA valuation adjustments, larger than CVA or FVA.

2.4.2. Initial Margin Valuation Adjustment (MVA)

MVA reflects the cost of funding of the initial margin requirements on top of other valuation adjustments. It is, too, being increasingly passed on to the end-client. Initial margin requirement is often based on Value-at-Risk (VaR) or Expected Shortfall (ES) approaches. To calculate MVA one must estimate expected profile of the initial margin over the life of the trade or portfolio of trades.

MVA is charged to the counterparty and modifies the equation (2.11) as follows

$$V = V_{risk-free} - CVA + DVA - FVA - KVA - MVA \tag{2.12}$$

To calculate the expected profile of the initial margin one needs to simulate multiple VaR/ES scenarios for each future date, which is computationally intensive. Green and Kenyon (2015) propose a more efficient approach using specific regression technique that allows fast recalculation of the portfolio value. In fact, Basel Committee and International Organization of Securities Commissions (Basel/IOSCO (2015)) published a revised framework for margin requirements for OTC derivatives with the objective of reducing systemic risk and motivating market participants to move to central clearing. The framework requires initial and variation margin to be exchanged between counterparties and puts limits on thresholds of these margins. Only institutions with over €8 billion of notional outstanding of OTC derivatives will be subject to initial margin requirements, and the largest derivatives dealers (notional outstanding over €3 trillion) will have to start posting initial margin for new transactions from September 2016 (smaller dealers later on and the requirements will be phased-in). If MVA adjustments are then passed on to end customers and are sizable, then it may have a significant impact on pricing and liquidity of derivatives.

3. Valuation of interest rate derivatives

In this chapter, we discuss discounting future cash flows of interest rate derivatives in detail. We start from the definition that the present value of cash flows of one leg of the swap equals the present value of cash flows of the other leg of the swap. OIS discounting takes the centre stage but we discuss other discount curves, too. The aim of this chapter is to derive discount factors and forward rates that are consistent with market prices. It is no longer plausible to use one curve to both estimate the forward rates and discount future cash flows given the prerequisite of arbitrage free market. We focus in detail on valuation of interest rate swaps, basis swaps and cross-currency swaps but first, we review relevant existing literature.

One of the first pre-crisis papers that address the issue of collateralization of derivatives is Johannes and Sundaresan (2003). The authors correctly note that collateralization as a way to eliminate counterparty credit risk bears additional costs and that the rate of the collateral impacts discount factors. In Johannes and Sundaresan (2007), the authors point to the impact of collateralization on swap rates and introduce formula to price collateralized contracts. They also show that if the collateral rate is continuous, then the rate of the collateral should be used to discount future cash flows even if such rate is not equal to the risk-free rate. Authors further present an empirical study of the US dollar swap rates dynamics. Many papers that followed refer to their ideas.

Pulman (2003), as one of the first authors, warns that it is not suitable to construct the yield curve from deposit, FRA/futures and swap rates. The author focuses on the short-end of the curve and shows that, for example, by using 1-month deposit rate to construct a 3-month swap curve (fixed swap rates against floating 3-month IBOR rates) one neglects the lower credit risk in the 1-month rate. Ametrano and Bianchetti (2009) introduce a model that focuses on segmentation of the market and uses the bootstrapping method to derive swap rates for each tenor respectively. The authors thus assume multiple discounting curves in one currency, this however does not eliminate arbitrage. Authors also ignore collateralization and cross-currency effects.

The paper of Boenkost and Schmidt (2005) is among the first to clearly focus on valuation of interest rate and cross-currency swaps including the effect of basis spreads and authors introduce the idea of two discount factors. Kijima et al. (2009) derive consistent pricing of USD/JPY cross-currency swaps, where they consider USD LIBOR as risk-free rate and calculate the JPY discount rate. Chibane et al. (2009) highlight the existence of basis spreads and show that valuation using one curve creates arbitrage opportunities between interest rate and currency swaps. Therefore they introduce new method of arbitrage-free pricing incorporating

basis spreads. Authors work only with IBOR-based swap rates and do not consider collateralization, or counterparty credit risk.

Fujii et al. (2010) extend the work of Johannes and Sundaresan (2007) for swaps where collateral is posted in different currency and construct arbitrage-free arguments for collateralized and uncollateralized interest rate and cross-currency swaps including quoted basis swap spreads. Mercurio (2010) suggests constructing discount curves using OIS swap rates, similar to Fujii et al. (2010). The impact of collateralization on derivatives pricing is discussed also in Piterbarg (2010) where the author modifies Black-Scholes valuation formula for collateralized options.

Hull and White (2013) consider OIS rates as currently the best proxy of risk-free rates and argue that they should be used as risk-free rates for valuation of all derivatives under risk-neutral measure. For completeness we list other relevant studies: Chibane (2012), Henrard (2009), Morini (2009), Tuckman and Porfirio (2003), Kijima et al. (2009).

3.1. Interest Rate Swap

An interest rate swap is a contract in which one party exchanges regular fixed interest rate cash flows in return for floating interest rate cash flows tied to a respective index, i.e. IBOR rate. The market value of an interest rate swap is thus the difference in present values of future cash flows of the fixed and floating rate leg. As for any other swap, at inception the present value of the fixed leg equals the present value of the floating leg

$$\sum_{t=1}^{n} f_t \delta_{1,t} DF_{d,t} = r_m \sum_{t=1}^{m} \delta_{2,t} DF_{d,t}, \tag{3.1}$$

where f_t are implied forward reference rates of the floating leg at times *t-1* to *t*, $\delta_{i,t} = \frac{D_t}{B}$ is the year fraction for period between *t-1* and *t* for *i*-th leg (i=1,2), D_t is the actual number of days between *t-1* and *t*, *B* is the number of days in a year according to the day count convention for the respective currency (e.g. for EUR, USD or CZK *B*=360), r_m is the quoted swap rate and $DF_{d,t}$ are discount factors (present values of one unit payment that will be received in time *t*) and usually there are different interest rate periods (i.e. *m*≠*n*).

The value of the floating leg depends on discount factors $DF_{d,t}$ and implied forward rates f_t. The swap rate that makes the net present value equal zero is then derived as

$$r_m = \frac{\sum_{t=1}^{n} f_t \delta_{1,t} DF_{d,t}}{\sum_{t=1}^{m} \delta_{2,t} DF_{d,t}}. \tag{3.2}$$

With given swap rates r_m and discount factors $DF_{d,t}$, we can calculate respective implied forward rates. In the following text, we discuss how to determine the discount curves (discount factors $DF_{d,t}$ that are used to calculate present value of future cash flows).

3.1.1. Classical bootstrapping using discount factors

A derivative position is valued using zero coupon yield curves. Movements in interest rates are reflected in discount factors that are used to obtain the present value of future derivative cash flows. The discount factors are derived recursively by what is called a bootstrapping process.

Discount factors (and the time structure of spot interest rates) are derived from available market prices of instruments in money market and swap market. First, we derive the initial discount factor for the shortest (overnight) tenor

$$DF_{d,ON} = \frac{1}{1+\delta_{ON} r_{ON}}, \tag{3.3}$$

where r_{ON} is the overnight (ON) money market cash rate. The remaining discount factors in the short end of the curve are derived consecutively from the shortest to the longest from already derived discount factors and money market and forward rate agreement (FRA) rates

$$DF_{d,TN} = \frac{1}{1+\delta_{TN} r_{TN}} DF_{d,ON}$$

$$DF_{d,t} = \frac{1}{1+\delta_t r_t} DF_{d,TN}\ , \tag{3.4}$$

where $DF_{d,TN}$ is discount factor for tomorrow next (tom-next, TN, meaning for one day starting tomorrow) money market rate. Every other money market deposit is settled in two business days (T+2).

Quoted swap rates are used to derive discount factors in the long end of the curve and the classical approach assumes that the present value of the swap's floating rate leg cash flows equals 100%.

$$1 = r_m \sum_{t=1}^{m} \delta_{2,t} DF_{d,t} + DF_{d,m}. \tag{3.5}$$

Using quoted swap rates r_m we can recursively calculate discount factors $DF_{d,t}$ from (3.5)

$$DF_{d,m} = \frac{1 - r_m \sum_{t=1}^{m-1} \delta_{2,t} DF_{d,t}}{1+\delta_{2,m} r_m}. \tag{3.6}$$

Here we assumed that forward rates f_t in (3.1) are implied from discount factors $DF_{d,t}$

$$f_t = \frac{1}{\delta_{1,t}}\left(\frac{DF_{d,t-1}}{DF_{d,t}} - 1\right). \tag{3.7}$$

After obtaining a set of discount factors from available specific market data, we need to use some form of interpolation to get discount factors for all other maturities (i.e. at dates between two dates with known discount factors). We can either interpolate on discount factors or convert discount factors into zero coupon (spot) rates and interpolate on the spot rates. Widely used market practice includes exponential interpolation on the discount factors and cubic splines or linear interpolation on the zero coupon curve. For a review of interpolation methods, we refer to Hagan and West (2006).

From the discount curve we can then construct a zero coupon curve, a par swap curve and as already shown, forward curves.

Swap rate r_m which makes the net present value equal zero (*par swap rate*) is then

$$r_m = \frac{\sum_{t=1}^{n}\left(\frac{DF_{d,t-1}}{DF_{d,t}}-1\right)DF_{d,t}}{\sum_{t=1}^{m}\delta_{2,t}DF_{d,t}} = \frac{1-DF_{d,n}}{\sum_{t=1}^{m}\delta_{2,t}DF_{d,t}}. \tag{3.8}$$

Equation (3.8) is a classical approach to valuation of interest rate swaps where we use one curve both to calculate forward rates and to discount future cash flows.

This approach can be found perhaps in almost every textbook on financial mathematics. It is based on the assumption that forward rates are implied from discount factors $DF_{d,t}$. This assumption is however not valid in the case when $DF_{d,t}$ are calculated from the risk-free rates. Then from (3.7) forward rates f_t are also considered risk-free rates but they are in fact IBOR rates that include credit and liquidity risk of interbank market. Therefore also equation (3.5) does not hold. One can however recursively calculate forward rates implied from discount factors $DF_{d,t}$ and quoted swap rates r_m from (3.1) as

$$f_n = \frac{r_m \sum_{t=1}^{m}\delta_{2,t}DF_{d,t} - \sum_{t=1}^{n-1} f_t\delta_{1,t}DF_{d,t}}{\delta_{1,n}DF_{d,n}} \tag{3.9}$$

For completeness we derive the last case when discount factors $DF_{d,t}$ and forward rates f_t are given. Then **swap rate** r_m which makes the net present value equal zero becomes

$$r_m = \frac{\sum_{t=1}^{n} f_t\delta_{1,t}DF_{d,t}}{\sum_{t=1}^{m}\delta_{2,t}DF_{d,t}}. \tag{3.10}$$

Therefore we always need two sets of factors from r_m, $DF_{d,t}$, and f_t to calculate the third one.

In the following section we focus on determining risk-free discount factors $DF_{d,t}$ and we further develop the concept of multiple curves.

3.1.2. OIS rates, collateralization, and OIS discounting

Overnight indexed swap (OIS) is an interest rate swap in which one party exchanges fixed rate for floating rate, which equals geometric average of overnight ONIA[3] rates during the life of the contract. ONIA rates reflect weighted average of market rates of unsecured overnight (O/N) deposits and loans transacted by reference banks in the respective interbank market, where the weights are sizes of individual transactions. These overnight rates are in close relationship with central banks' deposit rates, which are used to direct evolution of interest rates in the real economy.

Market participants can thus use OIS swaps to hedge against changes in central banks' rates, and vice versa, OIS swap rates reflect market expectation about central banks' rates. For example, if OIS rates are lower than current overnight rates then market signalizes incoming easing of monetary policy. These market expectations are, of course, priced in other interest rate instruments (FRAs, futures, interest rate swaps) but these include higher credit and liquidity risk and can thus distort information about expected central bank rates. OIS rates can be also distorted during times of low liquidity and at that time it can be problematic to use them for estimating future monetary policy steps. However, the underlying reference rate in OIS swaps is based on actual market transactions; therefore, OIS swaps better reflect market conditions and expectations.

Derivatives in OTC markets are commonly transacted and cleared bilaterally under ISDA[4] master agreement. The decrease in confidence and trust in the interbank markets led banks to search for ways to mitigate credit risk exposure from derivatives transactions. Today, it is common practice that part of ISDA agreement is the CSA annex[5], which specifies rules and conditions for credit support and collateral for mutual derivative transactions. It also agrees on steps to be taken in case of failure to make the required payment, posting a collateral, or a default of either of the counterparty. Derivatives are thus *collateralized*; they are secured by liquid financial collateral that offsets the credit exposure of one party vs. the other. A party that is *in the money* thus receives the collateral.

[3] ONIA – OverNight Index Average, i.e. EONIA for EUR, Federal funds effective rate for USD and CZEONIA (Czech OverNight Index Average) for CZK.
[4] ISDA - International Swaps and Derivatives Association master agreement, Czech banks also use Czech *master agreement for financial transactions* (CMA) published by Czech banking association and its respective product annex no. 5 for derivative transactions.
[5] CSA – Credit support annex, in case of CMA it is the product annex no. 11 *Margin maintenance annex.*

According to ISDA Margin Survey 2015 (ISDA (2015)), the use of collateral is extensive: in 2014 89% of non-cleared fixed-income derivatives and 97% of non-cleared credit derivatives were collateralized with CSA agreement, more than 80% of derivatives portfolios with more than 2500 trades are reconciled daily, and the use of cash or government securities accounts for over 90% of total collateral. These numbers have had an increasing trend over past years.

OTC derivatives market is gradually moving from bilateral clearing to clearing via central counterparty (CCP), which steps between the buyer and the seller and becomes counterparty to each of them. CCP then introduces additional techniques to mitigate and monitor counterparty credit risk (initial and variation margins, default fund contributions). When one of the members fails to post the required collateral, the member defaults and his positions are closed out.

The role of a CCP is to improve management of counterparty credit risk and increase transparency (availability of data). New regulation for both bilaterally or centrally cleared derivatives will require banks to post higher amounts of collateral, mainly in cash or liquid high quality securities. In this sense, a shift from managing counterparty credit risk towards managing liquidity risk of collateral outflow can be expected, see Finger (2012). By estimating short term volatility of market value of open derivative contracts or for different stress scenarios, one can estimate expenses for additional liquidity outflow (in the form of additional margin call sent to a CCP), i.e. by a quantile based models. As we discuss below, due to daily cash margining, CCPs have adopted OIS discounting.

As we discussed in Chapter 2 different valuation adjustment that were mainly focused on the uncollateralized part of the derivative, it may happen that derivatives prices may be different also when using central counterparties. In other words, it may happen that the same derivative will have a different price if it is being cleared at a different central counterparty, and this price would again depend on the collateralization rules being applied.

EU and the central counterparties

In 2009, G-20 leaders agreed to regulate OTC derivatives to decrease their credit and operational risks. The regulation deals mainly with derivatives trading, clearing, and reporting. To meet this agreement, the European Commission proposed and the Council of the EU and the European Parliament then adopted following legislation:

- On 4 July 2012, a *Regulation on OTC derivatives, central counterparties and trade repositories*[6], which specifies requirements for clearing and reporting for OTC

[6] EMIR - European Market Infrastructure Regulation: http://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32012R0648&from=EN

derivatives and requirements for activities of central counterparties and trade repositories.

- On 19 December 2012[7], *regulatory technical standards* and details on their implementation which among other things further specify requirements for central counterparties (including capital requirement), clearing obligation, indirect clearing arrangements, risk mitigation techniques for derivatives not cleared by a CCP, and data to be reported to trade repositories.
- On 6 August 2015[8], *a Regulation on central clearing for interest rate derivatives* that make central clearing of interest rate derivatives mandatory. It covers plain vanilla interest rate swaps, FRAs, OIS swaps and basis swaps in major currencies.

OIS discounting

Using OIS curve to discount derivative cash flows gained a lot of traction in both academia and among market professionals. In fact, discounting cash flows with OIS rates has become a market standard. Collateral used is most frequently cash which is accrued at overnight ONIA rate and collateralized portfolios are revalued daily, that is, margin call is calculated every day. With perfect hedging, collateral value is equal to market value of swap at maturity *T*, that is, the net interest payment at time *T*. The overnight rate accrued on the collateral is traded in the OIS swap market and is equivalent to fixed OIS rate for respective maturity. The value of the collateral and the market value of the interest rate swap are equal only if interest payments are discounted with OIS curve. For these reasons the valuation of derivatives has changed, and both banks and central counterparties started discounting at the OIS curve. One of the direct implications is the separation of discounting curve (OIS swaps) and forwarding curve (forward rates derived from the quoted fixed swap rates vs. IBOR) and as we show below, the implied forward rates are not the same if we use classical IBOR discounting or if we use OIS discounting. For more detailed discussion regarding valuation of collateralized trades and the choice of the risk free curve, we refer to Piterbarg (2010) and Fujii et al. (2010).

The floating rate *r* of the OIS swap is calculated as annual effective interest rate from the overnight ONIA rates for respective interest period

$$r = \left(\prod_{t=1}^{n_b}\left(1+\frac{r_t \times n_t}{360}\right)-1\right)\frac{360}{n},$$

where r_t is the overnight ONIA rate valid at time t and for a given currency, n_t is a number of days for which r_t is valid (usually $n_t = 1$, during weekends $n_t = 3$ etc.), n_b is number of business days for the given period (that is, number of rates r_t) and *n* is the total number of days

[7] Press release: http://europa.eu/rapid/press-release_IP-12-1419_en.htm?locale=en
[8] Draft: http://ec.europa.eu/finance/financial-markets/docs/derivatives/150806-delegated-act_en.pdf

in the interest rate period[9]. In other words, the floating rate payment equals swap notional repeatedly invested at overnight ONIA rate (compounded interest).

From quoted OIS swap rates r_{OIS} we can calculate discount factors; for one period (one fixed vs. floating payment) we can write

$$DF_{OIS,1} = \frac{1}{1 + \delta_{2,1} r_{OIS,1}}.$$

For multiple periods we proceed analogically as in (3.6)

$$DF_{OIS,n} = \frac{1 - r_{OIS,n} \sum_{t=1}^{n-1} \delta_{2,t} DF_{OIS,t}}{1 + \delta_{2,n} r_{OIS,n}}. \tag{3.11}$$

We use single OIS curve to derive the discount factors and implied forward OIS rates. The implied forward OIS rate $f_{OIS,t}$ and OIS *par swap rate* $r_{OIS,m}$ then become

$$f_{OIS,n} = \frac{r_{OIS,m} \sum_{t=1}^{m} \delta_{2,t} DF_{OIS,t} - \sum_{t=1}^{n-1} f_{OIS,t} \delta_{1,t} DF_{OIS,t}}{\delta_{1,n} DF_{OIS,n}} = \frac{1}{\delta_{1,n}} \left(\frac{DF_{OIS,n-1}}{DF_{OIS,n}} - 1 \right),$$

$$r_{OIS,m} = \frac{\sum_{t=1}^{n} f_{OIS,t} \delta_{1,t} DF_{OIS,t}}{\sum_{t=1}^{m} \delta_{2,t} DF_{OIS,t}} = \frac{1 - DF_{OIS,n}}{\sum_{t=1}^{m} \delta_{2,t} DF_{OIS,t}}. \tag{3.12}$$

3.1.3. Collateralized interest rate swap

Here we restrict ourselves to collateralized interest rate swap with zero thresholds, daily margining, and symmetric cash collateral which pays the ONIA rate. In such case, OIS rates should be used to discount future cash flows. If quoted swap rates are collateralized, then we derive implied forward rates from quoted swap rates and already derived OIS discount factors. Using OIS discounting, implied forward rates become

$$f_n = \frac{r_m \sum_{t=1}^{m} \delta_{2,t} DF_{OIS,t} - \sum_{t=1}^{n-1} f_t \delta_{1,t} DF_{OIS,t}}{\delta_{1,n} DF_{OIS,n}}. \tag{3.13}$$

One of the implications of changing the discounting curve is that the implied forward rate f_n of the collateralized swap is different than in (3.7). The swap rate r_m, which makes the NPV zero at inception, is then equal to

$$r_m = \frac{\sum_{t=1}^{n} f_t \delta_{1,t} DF_{OIS,t}}{\sum_{t=1}^{m} \delta_{2,t} DF_{OIS,t}}. \tag{3.14}$$

[9] Number of days in a year is given by money market day count convention for the respective currency.

3.1.4. Discount curve defined by spread to a reference rate

We can generalize previous cases by using implied forward reference rates f_t plus a spread s_t (IBOR + spread) to discount future cash flows. An arbitrage-free relation must hold for such unsecured financing

$$1 = \sum_{t=1}^{n} \delta_{1,t}(f_t + s_t)DF_{d,t} + DF_{d,n} \tag{3.15}$$

By using $f_t + s_t$ we treat all cases in the same manner; for example, for OIS discounting we set $s_t = r_{OIS,n} - r_n$ (which can be negative) and we get (3.11). However, we need a quoted swap market with reference rate f_t to determine f_t and $DF_{d,t}$

$$\sum_{t=1}^{n} f_t \delta_{1,t} DF_{d,t} = r_{n,s} \sum_{t=1}^{n} \delta_{2,t} DF_{d,t}. \tag{3.16}$$

Such market exists only if other subjects reflect the same risk of $f_t + s_t$, what is not the case, therefore, $r_{n,s}$ need to be approximated by existing swap market, for example, quoted swap market r_n from (3.8). Then

$$\sum_{t=1}^{n} f_t \delta_{1,t} DF_{d,t} = r_n \sum_{t=1}^{n} \delta_{2,t} DF_{d,t} \tag{3.17}$$

We then obtain discount factors by plugging (3.17) into (3.15)

$$DF_{d,n} = \frac{1-(r_n+s_n)\sum_{t=1}^{n-1} \delta_{2,t} DF_{d,t}}{1+\delta_{2,n}(r_n+s_n)}, \tag{3.18}$$

And forward rates f_n

$$f_n = \frac{r_n \sum_{t=1}^{n} \delta_{2,t} DF_{d,t} - \sum_{t=1}^{n-1} f_t \delta_{1,t} DF_{d,t}}{\delta_{1,n} DF_{d,n}}. \tag{3.19}$$

One of the reasons to mention this generalization of discounting with $f_t + s_t$ is that the time structure of spreads s_t shows the spread of the discount curve to forward curve in a similar way that basis spreads determine the level of one forward curve to another one. However, such discount curve can only be used if both parties reflect the same risk over the life of the swap. This is very rarely the case. In fact, as we discussed in Chapter 2, what has become a standard market practice is that cash flows are discounted at a proxy of a risk-free rate (most commonly OIS rates or sometimes repo rates if government bonds are used as collateral) and risk premiums of each of the swap counterparty are separately added on top of that.

3.2. Basis swap

Basis swap is a single currency floating-floating interest rate swap with each leg having a different tenor, that is, basis swap exchanges regular payments linked to one index for payments linked to other index plus a spread. Before the crisis these spreads were in the range of few basis points (almost negligible), but the change in risk perception for individual tenors caused (generally) positive spreads, which reflect higher credit and liquidity risk connected to longer tenors (see Figure 2).

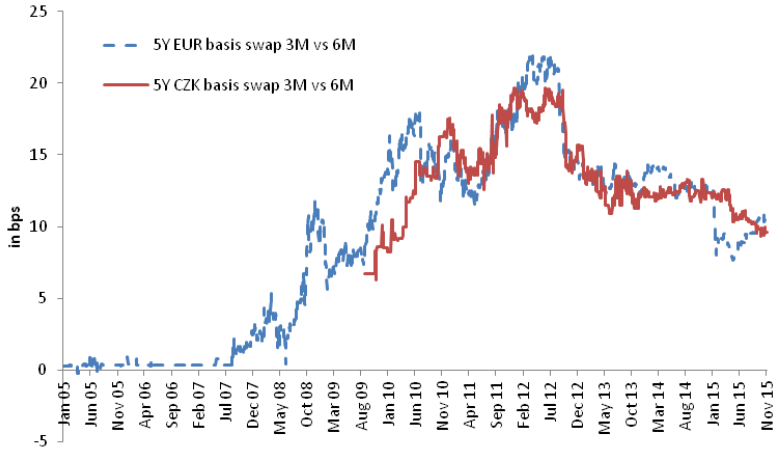


Figure 2. 5-year spreads in 3M/6M Euribor and 3M/6M Pribor basis swaps.
Source: Bloomberg

The present value of the first leg equals the present value of the second leg at inception

$$\sum_{t=1}^{n} f_{1,t}\delta_{1,t}DF_{d,t} = \sum_{t=1}^{m}\left(f_{2,t} + bs_m\right)\delta_{2,t}DF_{d,t}, \tag{3.20}$$

where $f_{i,t}$ are the implied forward reference rates and bs_m is basis swap spread, which can be positive or negative. With zero net present value (NPV) at inception this spread can be expressed as

$$bs_m = \frac{\sum_{t=1}^{n} f_{1,t}\delta_{1,t}DF_{d,t} - \sum_{t=1}^{m} f_{2,t}\delta_{2,t}DF_{d,t}}{\sum_{t=1}^{m}\delta_{2,t}DF_{d,t}} \tag{3.21}$$

Market value of the basis swap thus depends on three curves, i.e. with given discount curve and forward curves of one leg we can calculate forward rates of the other leg. The issue of basis swap spreads is well explained in Chibane et al. (2009) to where we refer for more detailed discussion.

For illustration only, in the first case we assume that discount factors $DF_{d,t}$ are calculated from swap rates with reference rate of the floating leg $f_{1,t}$ as in (3.1). Then we can write

$$r_{1,n}\sum_{t=1}^{n}\delta_{1,t}DF_{d,t} = \sum_{t=1}^{n} f_{1,t}\delta_{1,t}DF_{d,t} = 1 - DF_{d,n}$$

and we can calculate discount factors

$$DF_{d,n} = \frac{1 - r_{1,n}\sum_{t=1}^{n-1}\delta_{1,t}DF_{d,t}}{1 + \delta_{1,n}r_{1,n}},$$

forward rates $f_{1,t}$

$$f_{1,t} = \frac{1}{\delta_{1,t}}\left(\frac{DF_{d,t-1}}{DF_{d,t}} - 1\right),$$

and plugging this into (3.20) we calculate forward rates of the second leg $f_{2,m}$

$$f_{2,m} = \frac{1}{\delta_{2,m}DF_{d,m}}\left(r_{1,n}\sum_{t=1}^{n}\delta_{1,t}DF_{d,t} - bs_m\sum_{t=1}^{m}\delta_{2,t}DF_{d,t} - \sum_{t=1}^{m-1} f_{2,t}\delta_{2,t}DF_{d,t}\right). \quad (3.22)$$

3.2.1. Collateralized basis swap

Now we assume our central case, a basis swap as in (3.20), which is fully collateralized. Discount factors $DF_{OIS,t}$ are derived from the OIS swap curve as in (3.11). We also assume that swap rates r_n are exchanged against reference rate $f_{1,t}$, and to simplify notation we assume for the swap r_n same interest rate payment frequency for both fixed and floating leg (same $\delta_{1,t}$ for both fixed and floating leg). Then according to (3.13) we can write

$$f_{1,n} = \frac{r_n\sum_{t=1}^{n}\delta_{1,t}DF_{OIS,t} - \sum_{t=1}^{n-1} f_{1,t}\delta_{1,t}DF_{OIS,t}}{\delta_{1,n}DF_{OIS,n}},$$

where r_n is the quoted swap rate for maturity *n.* Now we can recursively calculate forward rates $f_{2,m}$ according to (3.22)

$$f_{2,m} = \frac{1}{\delta_{2,m}DF_{OIS,m}}\left(\sum_{t=1}^{n} f_{1,t}\delta_{1,t}DF_{OIS,t} - bs_m\sum_{t=1}^{m}\delta_{2,t}DF_{OIS,t} - \sum_{t=1}^{m-1} f_{2,t}\delta_{2,t}DF_{OIS,t}\right).$$

We see that basis spread bs_m enters the equation as an independent variable that cannot be omitted.

3.3. Cross-currency swap

Cross-currency swap (CCS) is an interest rate swap that exchanges periodic interest payments in different currencies and exchange rate used to fix initial and final notional is determined at inception. For more detailed explanation of the mechanism of cross-currency and basis swaps we refer the reader to Tuckman and Porfirio (2003).

A classical approach assumes that both legs of either cross-currency or basis swap trade without a spread, that is, it abstracts from credit and liquidity risks and assumes that each counterparty reflects the interbank risk of IBOR rates (and thus funds at swap rates), which was somewhat true until summer 2007. Present value of each leg of the swap was equal to its notional and the spread on one leg of the swap was in the range of several basis points (bps) what used to be neglected for valuation purposes. Today, the costs associated with funding in one currency versus funding in another are not negligible anymore. Temporary transfer of assets or liabilities in one currency into another currency poses additional credit and liquidity premium in the form of spread over one leg of the swap (Figure 3).

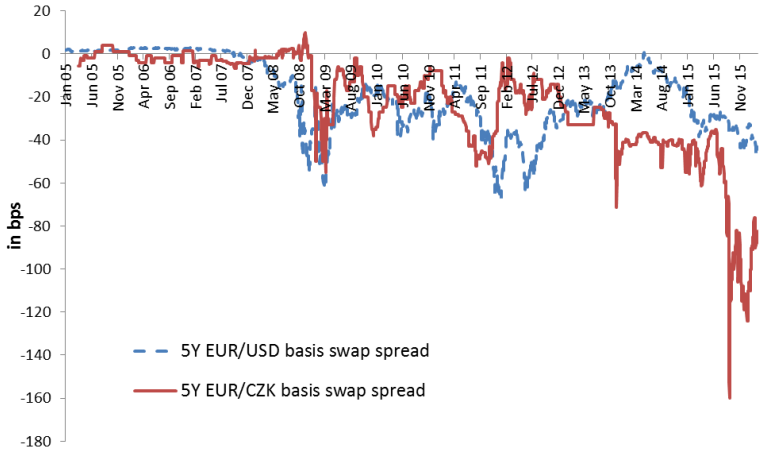


Figure 3. Five year EUR/USD and EUR/CZK cross-currency basis swap spread (3M USD LIBOR vs. Euribor + spread, 3M Euribor vs. 3M Pribor + spread) since 2005.
Source: Bloomberg

In Chapter 5 we discuss main drivers behind cross-currency basis swap spreads, such as credit and liquidity risks of the two underlying money market rates and supply and demand forces for one currency vs. another one.

Since a swap has zero NPV at inception (or close to zero, except for bid ask spread), it holds that present value of the first leg PV_1 is equal to present value of the second leg PV_2. When discounting swap leg with nonzero spread we use discount factors that take this quoted spread into account.

$$\sum_{t=1}^{n} f_{1,t}\,\delta_{1,t} DF_{1,t} + DF_{1,n} = \sum_{t=1}^{n} \left(f_{2,t} + BS_n\right) \delta_{2,t} DF_{2,t} + DF_{2,n}, \tag{3.23}$$

where $f_{j,t}$ is forward reference rate for period from *t-1* to *t*, $\delta_{j,t} = \frac{D_{j,t}}{B_j}$ is year fraction, $DF_{j,t}$ is the discount factor for currencies *j=1,2* and BS_n is the agreed basis spread (we expect that the spot fx rate equals the fx rate of the cross-currency swap at inception so we omit this term for simplicity). Discount factors $DF_{1,t}, t = 1, \dots, n$ are calculated so that the equality (3.23) holds. To simplify notation we assume that in (3.23) and further the frequency of interest payment is the same for both legs.

Again, for illustration only, let us assume that discount factors $DF_{1,t}$ are calculated from the swap rates with reference rate of the floating leg $f_{1,t}$ then the following equality holds

$$r_{1,n} \sum_{t=1}^{n} \delta_{1,t} DF_{1,t} = \sum_{t=1}^{n} f_{1,t}\,\delta_{1,t} DF_{1,t} = 1 - DF_{1,n}\ . \tag{3.24}$$

Plugging (3.24) into (3.23) yields

$$1 = \sum_{t=1}^{n} \left(f_{2,t} + BS_n\right) \delta_{2,t} DF_{2,t} + DF_{2,n}. \tag{3.25}$$

Similarly, we can write for the second leg

$$r_{2,n} \sum_{t=1}^{n} \delta_{2,t} DF_{2,t} = \sum_{t=1}^{n} f_{2,t}\,\delta_{2,t} DF_{2,t}. \tag{3.26}$$

Plugging (3.26) into (3.25) we can write

$$\left(r_{2,n} + BS_n\right) \sum_{t=1}^{n} \delta_{2,t} DF_{2,t} = 1 - DF_{2,n}, \tag{3.27}$$

from there we separate discount factors $DF_{2,t}$.

For the first period we have

$$DF_{2,1} = \frac{1}{1+\left(r_{2,1}+BS_1\right)\delta_{2,1}},$$

where $r_{2,1}$ is the spot rate for the first period (i.e. IBOR), BS_1 is the basis swap spread for the first period (i.e. 3 months) and $\delta_{2,1}$ is the first period's year fraction. Each following discount factor is calculated recursively from the previous discount factors using (3.27)

$$DF_{2,n} = \frac{1-\left(r_{2,n}+BS_n\right)\sum_{t=1}^{n-1} \delta_{2,t} DF_{2,t}}{1+\left(r_{2,n}+BS_n\right)\delta_{2,n}}. \tag{3.28}$$

Forward rates $f_{2,n}$ can be calculated from (3.26)

$$f_{2,n} = \frac{r_{2,n} \sum_{t=1}^{n} \delta_{2,t} DF_{2,t} - \sum_{t=1}^{n-1} f_{2,t} \delta_{2,t} DF_{2,t}}{\delta_{2,n} DF_{2,n}}. \tag{3.29}$$

To value cross-currency swap with nonzero basis spread *BS* we have to use two different curves, $DF_{1,n}$ for discounting the cash flows of the first leg and $DF_{2,n}$ for estimating $f_{2,n}$ and discounting the second leg's cash flows. Although assumption of discounting at swap rates does not hold anymore, we did so to illustrate how quoted basis spread impacts the calculation of discount factors $DF_{2,n}$ and $f_{2,n}$.

3.3.1. Collateralized cross-currency swap

We now assume full collateralization with overnight ONIA cash rate in the currency of the first leg and therefore we use this rate to discount cash flows of the first leg. The present value of the first leg equals the present value of the second leg, but that is not equal *par* any longer (forward rates of the first leg are calculated from the quoted swap rates and discounted with OIS curve, which is different and generally lower than the swap curve).

$$\sum_{t=1}^{n} f_{1,t}\,\delta_{1,t} DF_{OIS_1,t} + DF_{OIS_1,n} = \sum_{t=1}^{n} \left(f_{2,t} + BS_n\right) \delta_{2,t} DF_{2,t} + DF_{2,n}, \tag{3.30}$$

Discount factors $DF_{OIS_1,t}$ of the first leg are calculated from the OIS swap curve according to (3.11). We assume that market quotes fixed swap rate $r_{1,n}$ against floating rates $f_{1,n}$, then according to (3.13) we can write

$$f_{1,n} = \frac{r_{1,n} \sum_{t=1}^{n} \delta_{1,t} DF_{OIS_1,t} - \sum_{t=1}^{n-1} f_{1,t} \delta_{1,t} DF_{OIS_1,t}}{\delta_{1,n} DF_{OIS_1,n}}.$$

We also assume that market quotes fixed swap rate $r_{2,n}$ against floating rates $f_{2,n}$

$$r_{2,n} \sum_{t=1}^{n} \delta_{2,t} DF_{2,t} = \sum_{t=1}^{n} f_{2,t}\,\delta_{2,t} DF_{2,t}.$$

Plugging this into (3.30) we get

$$\sum_{t=1}^{n} f_{1,t}\,\delta_{1,t} DF_{OIS_1,t} + DF_{OIS_1,n} = \left(r_{2,n} + BS_n\right) \sum_{t=1}^{n} \delta_{2,t} DF_{2,t} + DF_{2,n},$$

and we can isolate discount factors $DF_{2,t}$.

For the first period we have

$$DF_{2,1} = \frac{\left(1 + r_{1,1}\delta_{1,1}\right) DF_{OIS_1,1}}{1 + \left(r_{2,1} + BS_1\right)\delta_{2,1}},$$

where $r_{1,1}$ is the spot rate for the first period (i.e. IBOR), $DF_{OIS_1,1}$ is the OIS discount factor for the first period, $\delta_{1,1}$ is the year fraction in day count convention of the first leg for the first interest rate period. For each following discount factor we can write

$$DF_{2,n} = \frac{\sum_{t=1}^{n} f_{1,t}\delta_{1,t}DF_{OIS_1,t} + DF_{OIS_1,n} - (r_{2,n} + BS_n)\sum_{t=1}^{n-1}\delta_{2,t}DF_{2,t}}{1 + (r_{2,n} + BS_n)\delta_{2,n}} \tag{3.31}$$

Basis swap spread changes discount factors $DF_{2,n}$ of the second leg. We also notice that the present value of neither leg in (3.30) equals *par.*

3.3.2. Cross-currency swap collateralized in the currency of the second leg

We assume that the cross currency swap is now fully collateralized in the currency of the second leg of the swap

$$\sum_{t=1}^{n} f_{1,t}\,\delta_{1,t}DF_{1,t} + DF_{1,n} = \sum_{t=1}^{n}\left(f_{2,t} + BS_n\right)\delta_{2,t}DF_{OIS_2,t} + DF_{OIS_2,n}. \tag{3.32}$$

We also assume that the OIS swap market of the currency of the second leg and the swap market with reference rates $f_{2,t}$ exist, so we can calculate discount factors $DF_{OIS_2,t}$ from (3.11) and forward rates $f_{2,t}$ from (3.13). We also need a swap market for $f_{1,t}$

$$r_{1,n}\sum_{t=1}^{n}\delta_{1,t}DF_{1,t} = \sum_{t=1}^{n} f_{1,t}\,\delta_{1,t}DF_{1,t}.$$

Plugging this into (3.32) we get

$$r_{1,n}\sum_{t=1}^{n}\delta_{1,t}DF_{1,t} + DF_{1,n} = \sum_{t=1}^{n}\left(f_{2,t} + BS_n\right)\delta_{2,t}DF_{OIS_2,t} + DF_{OIS_2,n},$$

And we can isolate discount factors $DF_{1,n}$

$$DF_{1,n} = \frac{\sum_{t=1}^{n}(f_{2,t} + BS_n)\delta_{2,t}DF_{OIS_2,t} + DF_{OIS_2,n} - r_{1,n}\sum_{t=1}^{n-1}\delta_{1,t}DF_{1,t}}{1 + \delta_{1,n}r_{1,n}}. \tag{3.33}$$

Changing the currency of the collateral was the last case we wanted to illustrate. We have shown how to incorporate basis spreads into the valuation of basis swaps and cross-currency basis swaps, and we looked at the impact of collateralization on the discount factors and forward rates. In other words, we have looked into how to price interest rate, basis, and cross-currency swaps consistently with market quotes. We first illustrated the approach for the classical case of uncollateralized IBOR discounting but we mainly focused on collateralized derivatives (cash collateralization using OIS discounting).

For uncollateralized trades we also showed a general case of incorporating risk spreads into the calculation and discounting, and for collateralized trades, we discussed the impact of the currency of the collateral in case of cross-currency swaps.

We have seen that in the absence of arbitrage it is no longer possible to use one curve to both estimate forward rates and to discount future cash flows and we have introduced the concept of multiple curves.

3.4. Approximation of discount curves of cross-currency swaps

In this section we derive discount curves implied from cross-currency market quotes and the discount curve of the base currency. We follow Wu (2011) who explored relationship between discount curve implied from CCS basis spread quotes and LIBOR and OIS curves. His findings offer several interesting practical applications, for example, the discount curve of the second leg of the CCS can be approximated from both CCS IBOR basis swap and CCS OIS basis swap, or that the decomposition of CCS spread into OIS basis spread and spread reflecting credit and liquidity risk between two currencies can have a direct implication on the choice of currency for collateral posting.

The idea of such approximation is to express basis spread in terms of swap rates used to calculate forwards rates, and swap rates used to discount cash flows. Let $r_{j,f}$ be the quoted swap rate of *j*-th leg (*j=1,2*) for maturity *n,* from which the forward IBOR rates are derived and let $r_{j,d}$ be the swap rate of *j*-th leg corresponding to discount factors $DF_{j,t}$ (for example, OIS discount factors). If the day count convention and interest payment frequency is the same, then according to (3.1) and (3.8) we can write

$$\sum_{t=1}^{n} f_{j,t}\delta_{j,t}DF_{j,t} = r_{j,f}\sum_{t=1}^{n}\delta_{j,t}DF_{j,t}, r_{j,f} = \frac{\sum_{t=1}^{n} f_{j,t}\delta_{j,t}DF_{j,t}}{\sum_{t=1}^{n}\delta_{j,t}DF_{j,t}}, \quad (3.34)$$

$$r_{j,d}\sum_{t=1}^{n}\delta_{j,t}DF_{j,t} + DF_{j,n} = 1, r_{j,d} = \frac{1-DF_{j,n}}{\sum_{t=1}^{n}\delta_{j,t}DF_{j,t}}. \quad (3.35)$$

Substituting (3.34) and (3.35) into (3.23) we get

$$\left(r_{1,f} - r_{1,d}\right)\sum_{t=1}^{n}\delta_{1,t}DF_{1,t} + 1 = \left(r_{2,f} - r_{2,d} + BS_n\right)\sum_{t=1}^{n}\delta_{2,t}DF_{2,t} + 1, \quad (3.36)$$

And we can express BS_n as

$$\left(r_{1,f} - r_{1,d}\right)\frac{\sum_{t=1}^{n}\delta_{1,t}DF_{1,t}}{\sum_{t=1}^{n}\delta_{2,t}DF_{2,t}} = \left(r_{2,f} - r_{2,d} + BS_n\right)$$

$$BS_n = \left(r_{1,f} - r_{1,d}\right)\frac{\sum_{t=1}^{n}\delta_{1,t}DF_{1,t}}{\sum_{t=1}^{n}\delta_{2,t}DF_{2,t}} - \left(r_{2,f} - r_{2,d}\right) \quad (3.37)$$

The equation (3.37) will serve as a basis of the analysis as we further decompose basis spread BS_n into its credit and liquidity element $BS_{OIS-OIS}$ and supply and demand element $BS_{FX,OIS}$. In the following section we also use (3.37) to derive an approximate formula for selected types of cross-currency basis swaps according to Wu (2011).

3.4.1. Credit and liquidity risks between two currencies

When we compare present values of two collateralized (OIS discounted) floating legs in two different currencies we can approximate credit and liquidity risk of the two currencies $BS_{OIS-OIS}$, or credit and liquidity risk of one currency vs. another one.

$$\sum_{t=1}^{n} f_{1,t}\delta_{1,t}DF_{OIS_1,t} + DF_{OIS_1,n} = \sum_{t=1}^{n}\left(f_{2,t} + BS_{OIS-OIS}\right)\delta_{2,t}DF_{OIS_2,t} + DF_{OIS_2,n}.$$

Combining (3.37), (3.34), and (3.35) we get

$$BS_{OIS-OIS} = \left(r_{1,IBOR} - r_{1,OIS}\right)\frac{\sum_{t=1}^{n}\delta_{1,t}DF_{OIS_1,t}}{\sum_{t=1}^{n}\delta_{2,t}DF_{OIS_2,t}} - \left(r_{2,IBOR} - r_{2,OIS}\right),$$

where

$$r_{j,IBOR} = \frac{\sum_{t=1}^{n} f_{j,t}\delta_{j,t}DF_{OIS_j,t}}{\sum_{t=1}^{n}\delta_{j,t}DF_{OIS_j,t}},\ r_{j,OIS} = \frac{1-DF_{OIS_j,n}}{\sum_{t=1}^{n}\delta_{j,t}DF_{OIS_j,t}}$$

By rounding the ratio $\frac{\sum_{t=1}^{n}\delta_{1,t}DF_{OIS_1,t}}{\sum_{t=1}^{n}\delta_{2,t}DF_{OIS_2,t}}$ to 1 we get an approximate relationship for spread $BS_{OIS-OIS}$

$$BS_{OIS-OIS} \approx \left(r_{1,IBOR} - r_{1,OIS}\right) - \left(r_{2,IBOR} - r_{2,OIS}\right). \quad (3.38)$$

Cross-currency credit and liquidity risk can thus be approximated as the difference between interbank risk (IBOR and OIS spreads) in two currencies.

Likewise we can approximate swap rates used for discounting $r_{2,d} = r_{2,FX}$ implied from quoted IBOR-based basis spreads BS_{FX} and from $r_{1,OIS}$

$$BS_{FX} \approx \left(r_{1,IBOR} - r_{1,OIS}\right) - \left(r_{2,IBOR} - r_{2,FX}\right)$$

$$\begin{aligned} r_{2,FX} &\approx BS_{FX} + r_{2,IBOR} - \left(r_{1,IBOR} - r_{1,OIS}\right) \\ &\approx r_{2,OIS} + (BS_{FX} - BS_{OIS-OIS}). \end{aligned} \quad (3.39)$$

Swap rates $r_{2,FX}$ implied from basis spreads can thus be approximated as IBOR swap rates shifted by basis spread and reduced by IBOR-OIS spread of the first leg, or alternatively, as OIS rates shifted by factor $(BS_{FX} - BS_{OIS-OIS})$. It is worth noting that $r_{2,FX} \approx r_{2,OIS}$ only in case that $(BS_{FX} - BS_{OIS-OIS})$ is negligible. This means that it is not possible to estimate OIS swap rates of the second leg using only $r_{2,FX}$.

3.4.2. Cross-currency OIS basis swap (float-float)

Floating to floating cross-currency OIS basis swap temporarily transfers liquidity in one currency in exchange for liquidity in another currency. During the life of the trade one party pays periodic interest payments linked to overnight reference rates (ONIA) plus a spread (payments occur for example every 3 months) and receives interest payments linked to overnight reference rates in another currency[10].

For those currencies for which OIS cross-currency market exists, we can estimate swap rates used for discounting in a similar fashion as in (3.39)

$$\sum_{t=1}^{n} f_{OIS_1,t}\delta_{1,t}DF_{OIS_1,t} + DF_{OIS_1,n} = \sum_{t=1}^{n}\left(f_{OIS_2,t} + BS_{FX,OIS}\right)\delta_{2,t}DF_{FX,t} + DF_{FX,n}, \quad (3.40)$$

where forward rates $f_{OIS_j,t} = \frac{1}{\delta_{j,t}}\left(\frac{DF_{OIS_j,t-1}}{DF_{OIS_j,t}} - 1\right)$ are calculated from OIS discount factors and therefore the left side equals 1. Using (3.38) and (3.39) we get an approximation

$$\begin{aligned} BS_{FX,OIS} &\approx -\left(r_{2,OIS} - r_{2,FX}\right) \\ &\approx BS_{FX} + \left(r_{2,IBOR} - r_{2,OIS}\right) - \left(r_{1,IBOR} - r_{1,OIS}\right) \\ &\approx BS_{FX} - BS_{OIS-OIS}. \end{aligned} \quad (3.41)$$

OIS basis spread $BS_{FX,OIS}$ therefore represents the difference between the discount rate implied from discount factors of the second leg and market OIS rate of the second leg and can be approximated as IBOR basis spread BS_{FX} reduced by basis spread $BS_{OIS-OIS}$ that reflect credit and liquidity risk between the two currencies.

[10] For example, in case of EUR/USD it is a basis swap EUR EONIA + spread vs. Fed funds effective rate.

We assumed that discount factors $DF_{FX,n}$ in (3.40) implied from the first leg of the cross-currency OIS swap are the same as in cross-currency swap in (3.30). We can verify this empirically for the case of EUR/USD: in the following graph we plot EUR/USD OIS basis swap curve (EONIA + spread vs Fed funds) and its approximation from EUR/USD LIBOR basis swap curve as in (3.41) as at 7 November, 2015.

EUR-USD BS FF/EONIA	Approximation	Difference in bps
-35.5	-35.29	-0.2
-33.8	-33.8	0.1
-35.3	-35.35	0.1
-37.3	-37.775	0.5
-42.0	-42.3	0.3
-45.5	-46.675	1.2
-47.8	-48.325	0.6
-49.3	-48.925	-0.3
-51.0	-50.075	-0.9
-51.0	-49.65	-1.4
-47.8	-45.75	-2.0
-46.5	-44.25	-2.3
-45.8	-44.1	-1.7

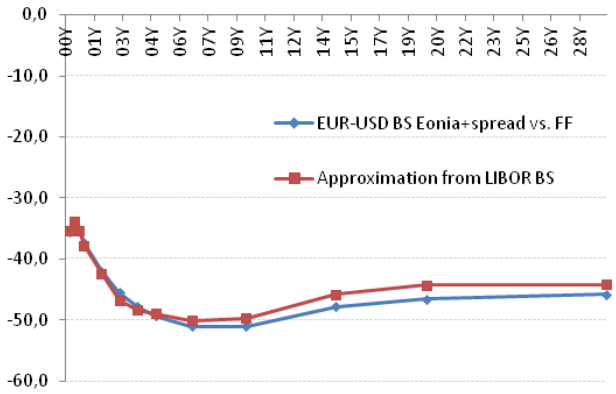


Table 1, Figure 4. Comparison of market quotes of EUR/USD OIS basis swap spreads and their approximations as at 7 November, 2015.
Source: Bloomberg, Author's calculations

Swap rates used for discounting $r_{2,FX}$ can be thus estimated from either the IBOR cross-currency basis swap curve or OIS cross-currency basis swap curve.

$$r_{2,FX} \approx BS_{FX} + r_{2,IBOR} - \left(r_{1,IBOR} - r_{1,OIS}\right) \approx r_{2,OIS} + BS_{FX,OIS}. \qquad (3.42)$$

Wu (2011) empirically showed that for liquid instruments (3.42) holds with accuracy of 1-3 basis points. One of the implications is that for discounting collateralized cross-currency swap we can thus use either the discount curve implied from IBOR basis spreads or the discount curve implied from OIS basis spreads.

For illustration, using the example of EUR/USD cross-currency swap, we will decompose (3.41) into the three underlying swaps, the cross-currency EUR/USD OIS swap, basis swap Eonia vs. 3M Euribor, and basis swap Fed funds vs. 3M USD Libor.

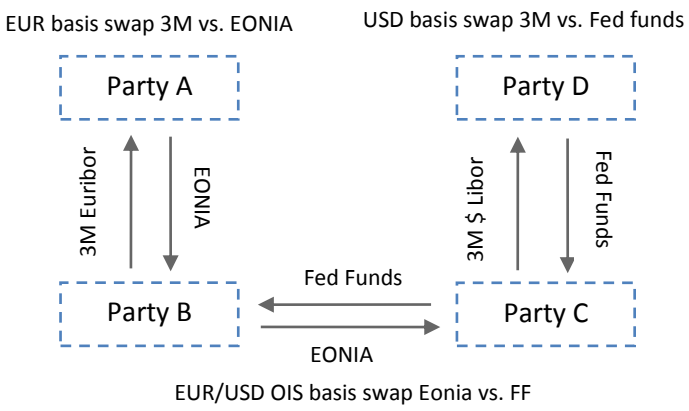


Figure 5. Decomposition of cross-currency EUR/USD IBOR swap.

Eonia vs. 3M Euribor swap is the $(r_{Euribor} - r_{Eonia})$ component and Fed funds vs. 3M USD Libor is the $(r_{USD\ Libor} - r_{USD\ OIS})$ component in (3.41). Basis spread BS_{FX} of the cross-currency swap therefore depends on the time structure of credit and liquidity spreads of domestic and foreign rates, and OIS basis.

3.4.3. Currency of collateral and indirect quotes

Generally, the currency of the derivative can be different from the currency of the collateral, or credit support annex allows collateral posting in different currencies (for example, USD interest rate swap collateralized by EUR cash). In this case, the party that is out of the money aims to optimize cash flows and deliver the cheapest collateral. Then it is suitable to discount future cash flows using cross-currency basis swap curves of the currency of the derivative and the currency of the collateral, i.e. using "cross-currency implied" OIS curve which corresponds to the OIS curve in collateral currency adjusted for OIS cross-currency basis spread.

This "new" OIS curve implied from cross-currency basis swap of currency 1 vs. currency 2 will be used to discount a derivative in currency 1 which is collateralized by cash in currency 2 according to (3.36)

$$\left(r_{1,IBOR} - r_{1,d}\right)\sum_{t=1}^{n}\delta_{1,t}DF_{1,t} = \left(r_{2,IBOR} - r_{2,OIS} + BS\right)\sum_{t=1}^{n}\delta_{2,t}DF_{2,t}.$$

$$r_{1,d} \approx r_{1,IBOR} - BS - \left(r_{2,IBOR} - r_{2,OIS}\right). \tag{3.43}$$

This is an approximate formula of (3.33).

For currency pairs without direct quotes, indirect quotes against a common currency need to be used (for example for USD/CZK cross-currency swap we will use quoted spreads of EUR/USD and EUR/CZK cross-currency swaps). We will use formula (3.36) and adjust the first leg for the quoted spread

$$\left(r_{1,IBOR} - r_{1,d} + BS_1\right)\sum_{t=1}^{n}\delta_{1,t}DF_{1,t} = \left(r_{2,IBOR} - r_{2,OIS} + BS_2\right)\sum_{t=1}^{n}\delta_{2,t}DF_{2,t},$$

$$r_{1,d} \approx r_{1,IBOR} + BS_1 - BS_2 - \left(r_{2,IBOR} - r_{2,OIS}\right). \tag{3.44}$$

We conclude this chapter by revisiting the simplifying assumption in (3.38) about equality of annuities $\sum_{t=1}^{n}\delta_{j,t}DF_{OIS_j,t}$ meaning that the two swap curves are close to each other and their discount factors have similar values. This may not hold for all currencies and especially not for longer maturities. Wu (2012) suggests a simple approximation assuming a flat curve: for a given swap rate $r_{j,n}$ we construct a flat curve whose discount factors form a geometric series

$$A_j = \sum_{t=1}^{n}\delta_{j,t}DF_{j,t} \approx \sum_{t=1}^{n}\bar{\delta}e^{-t\bar{\delta}r_t},$$

where $\bar{\delta}$ is the year fraction unadjusted for business days.

4. Cross-currency and OIS market in Czech koruna

In this chapter we discuss possible ways how to estimate OIS curve for the currency without the liquid OIS market in all tenors. We focus on Czech market that has only recently started to quote CZK OIS rates but results hold without loss of generality. We also decompose spread of the cross-currency swap into individual components that reflect credit and liquidity risk between the two currencies and relative demand of one currency versus the other. There is a straightforward application of such breakdown, it can be used to approximate implied discount rates of the second leg of the cross-currency swap and to estimate OIS rates in another currency (in our case CZK OIS rates, which reflect market expectations about central bank rates freed from credit and liquidity risk of Czech interbank rates). CZK OIS rates can be then used to discount derivatives denominated in CZK and collateralized with CZK cash.

4.1. CZK Interest rate swap

In case of CZK interest rate swap that is collateralized in CZK cash we can write according to (3.14)

$$\sum_{t=1}^{n} f_{Pribor,t}\delta_{1,t}DF_{CZK\ OIS,t} = r_{Pribor,n}\sum_{t=1}^{n}\delta_{2,t}DF_{CZK\ OIS,t}, \tag{4.1}$$

where $DF_{CZK\ OIS,t}$ are discount factors from CZK OIS rates $r_{OIS,t}$. CZK OIS rates are quoted up to 10 years but maturities above 1 year have started to be quoted only recently, since the 4Q 2013. For maturities above 10 years some form of approximation using available market data is needed.

4.2. Estimation of CZK OIS curve above 10 years

One possible way to estimate CZK OIS curve above 10 years is to define the discount curve by spread to reference rates as in (3.18). We set the reference rate as CZK swap rates and thus define the spread s_{10Y} as the difference between the 10-year swap rate $r_{Pribor,10Y}$ (fix vs. 3-month Pribor) and the 10-year OIS rate

$$s_{10Y} = r_{Pribor,10Y} - r_{OIS,10Y} \tag{4.2}$$

for maturities above 10 years. For example, as at 7 November 2015 $r_{Pribor,10Y} - r_{OIS,10Y} = 0.42\%$ (MID). We think that it is justifiable to assume that swap rates and OIS rates have a constant spread for maturities above 10 years. We can verify this assumption empirically for currencies with liquid OIS market. For example, the following graph shows the time structure of basis swap spreads of Fed Funds vs. 3M USD Libor and Eonia vs. 3M Euribor

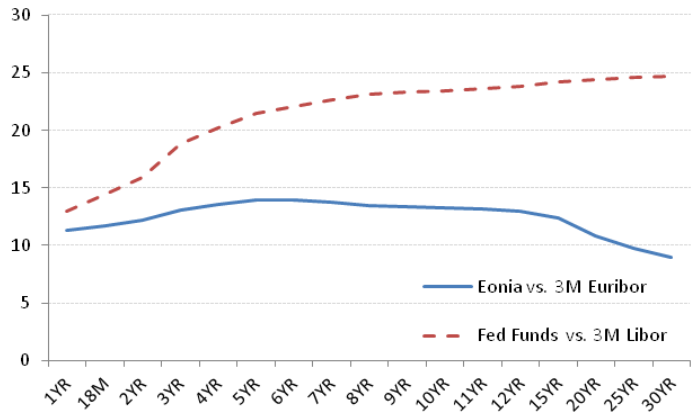


Figure 6. Time structure of basis swap spreads Fed Funds vs. 3M USD Libor and Eonia vs. 3M Euribor as at 7 November, 2015.
Source: Bloomberg

We can see that spreads stabilize above 4 years, becoming almost constant, with the exception of the very long end in case of euro swap market. The short end is usually more liquid and spread in the short end reflects market expectations about central bank policy interest rates. On top of that, swap rates also include market expectations of credit and liquidity risks in IBOR rates. These expectations are also mainly reflected in the short end of the curve, while in the long end it is more of an estimate of long-term level of policy rates plus an average of credit and liquidity risks in IBOR rates.

4.3. Approximation of discount rates of cross-currency swap

In the previous chapter we showed how it is possible to approximate swap rates, which determine discount factors of the second leg of the cross-currency swap, and which are implied from quotes of basis spreads. We did it by decomposing CCS spread into individual components reflecting risk between two currencies, but also relative demand of one currency vs. the other one. We can use this decomposition to estimate CZK OIS rates.

Using results from the previous chapter, we can approximate CCS basis spread as

$$BS_{FX} \approx \left(r_{1,IBOR} - r_{1,OIS}\right) - \left(r_{2,IBOR} - r_{2,FX}\right) \tag{4.3}$$

From (4.3) we can approximate swap rates used for discounting $r_{2,FX} = r_{2,d}$

$$\begin{aligned} r_{2,FX} &\approx BS_{FX} + r_{2,IBOR} - \left(r_{1,IBOR} - r_{1,OIS}\right) \\ &\approx r_{2,OIS} + (BS_{FX} - BS_{OIS-OIS}), \end{aligned} \tag{4.4}$$

where $r_{2,FX} = r_{2,d}$ and

$$r_{2,d} \sum_{t=1}^{n} \delta_{2,t} DF_{2,t} + DF_{2,n} = 1, r_{2,d} = \frac{1 - DF_{2,n}}{\sum_{t=1}^{n} \delta_{2,t} DF_{2,t}}.$$

Swap rates $r_{2,FX}$ can be approximated as IBOR rates shifted by basis spread and reduced by interbank risk of the first leg. Term $BS_{OIS-OIS}$ denotes the difference between interbank (credit and liquidity) risk of one currency vs. another one

$$BS_{OIS-OIS} \approx \left(r_{1,IBOR} - r_{1,OIS}\right) - \left(r_{2,IBOR} - r_{2,OIS}\right) \tag{4.5}$$

Similarly for OIS cross-currency swaps $BS_{FX,OIS}$ (cross-currency swaps that temporarily transfer liquidity with payments linked to overnight rates)

$$\begin{aligned} BS_{FX,OIS} &\approx -\left(r_{2,OIS} - r_{2,FX}\right) \\ &\approx BS_{FX} + \left(r_{2,IBOR} - r_{2,OIS}\right) - \left(r_{1,IBOR} - r_{1,OIS}\right) \\ &\approx BS_{FX} - BS_{OIS-OIS} \end{aligned} \tag{4.6}$$

By removing interbank risk from money market rates, we are left with OIS rates and the remaining spread in CCS OIS basis swap $BS_{FX,OIS}$ then reflects relative supply and demand of one currency vs. another one. We now use this approximation to estimate CZK OIS rates.

4.4. CZK discount curve implied from EUR/CZK cross-currency swap

If OIS basis spread $BS_{FX,OIS}$ in EUR/CZK cross-currency swap is negligible (meaning that basis spread between EUR and CZK currencies can be explained by credit and liquidity risk of one currency vs. another one), then CZK OIS discount curve can be approximated by (4.4) and $r_{2,FX} = r_{2,OIS}$. We can verify this empirically by comparing quoted CZK OIS rates with synthetic

rates from EUR/CZK FX swaps up to 1 year. We will use quoted EUR/CZK swap points and EONIA rates as inputs.

$$S + sp = S\frac{1 + r_{CZK\ OIS,t}\frac{D_t}{360}}{1 + r_{EONIA,t}\frac{D_t}{360}}$$

$$r_{CZK\ OIS,t} = \left[\frac{S+sp}{S}\left(1 + r_{EONIA,t}\frac{D_t}{360}\right) - 1\right]\frac{360}{D_t}, \tag{4.7}$$

where S is the EUR/CZK spot rate, sp are quoted swap points, D_t is the number of days of FX swap, $r_{EONIA,t}$ is the EUR OIS swap rate for maturity t, and $r_{CZK\ OIS,t}$ is the synthetic CZK OIS swap rate.

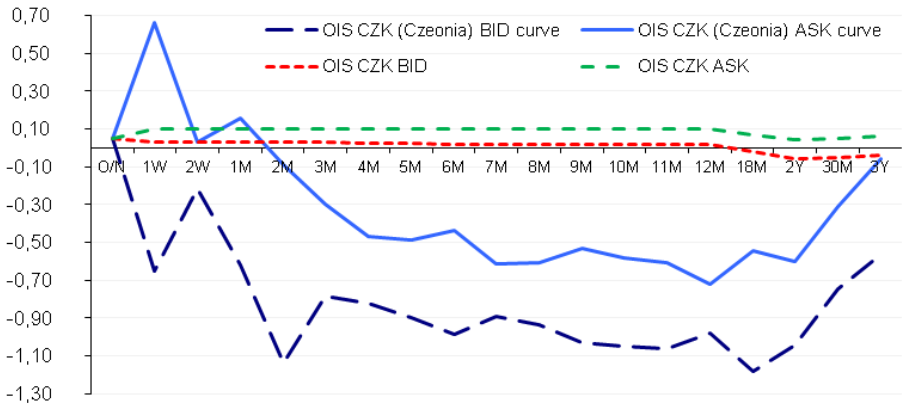


Figure 7. Comparing synthetic CZK OIS curve from FX swaps up to 3 years with quoted OIS market rates on 7 November, 2015.
Source: Bloomberg, Author's calculations

Synthetic rates from FX swaps do not correspond with quoted rates, in particular, during recent period. For example, on 7 November 2015, synthetic curve seems significantly dislocated and indicate a surplus of CZK liquidity which is pushing synthetic rates into deeper negative territory. This has not always been the case, for example during the course of 2012, synthetic CZK OIS rates were noticeably higher in the short end what indicated an increased demand for CZK liquidity via FX swaps, therefore $BS_{FX,OIS}$ was positive. In Chapter 5 we discuss how can market participant who has access to Czech national bank repo facility use this arbitrage opportunity by borrowing EUR, swapping EUR into CZK, and depositing CZK at the central bank.

If we use (4.2) to estimate CZK OIS rates above 10 years, then we can approximate $r_{CZK\ FX,t}$ swap rate implied from EUR/CZK basis swap spreads (Euribor 3M vs Pribor 3M + spread) with (4.4)

Maturity	r CZK FX MID	r CZK OIS
1 YR	-0.78	0.06
2 YR	-0.78	-0.01
3 YR	-0.66	0.01
4 YR	-0.51	0.06
5 YR	-0.36	0.11
6 YR	-0.23	0.17
7 YR	-0.10	0.25
8 YR	0.03	0.35
9 YR	0.16	0.44
10 YR	0.24	0.52
12 YR	0.42	0.55
15 YR	0.64	0.73
20 YR	0.84	0.87

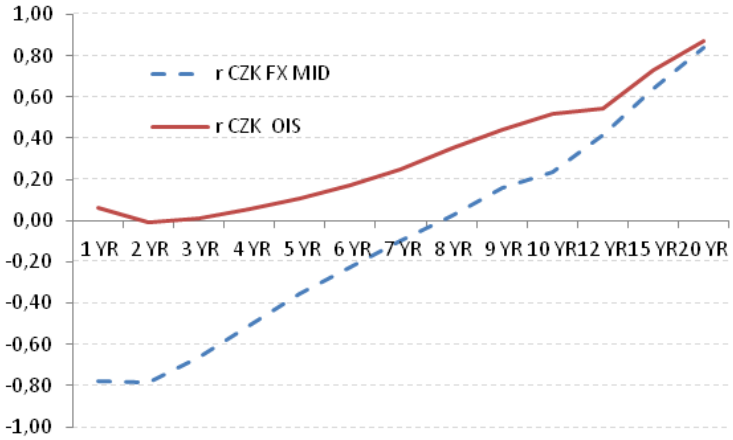


Table 2, Figure 8. Implied swap rates $r_{CZK\ FX,t}$ and $r_{CZK\ OIS,t}$ on 7 November 2015.
Source: Bloomberg, Author's calculations

Synthetic CZK rates are significantly below OIS CZK rates. This may have a practical impact on the valuation of derivatives. For example, let us assume that counterparties traded CZK interest rate swap under a CSA agreement that allows posting collateral in either CZK or EUR cash. Then a counterparty who is *out of the money* prefers to send CZK cash and thus discount future cash flows with higher rates $r_{CZK\ OIS,t}$ in order to achieve lower negative market value of the swap.

In the next section we decompose basis spread $BS_{EURCZK\ FX}$ to find whether it is the liquidity and credit risk premium of EUR rates that make synthetic CZK rates deeply negative or it is rather lower demand for CZK currency.

4.5. EUR/CZK Cross-currency OIS swap

We can use (4.6) to approximate basis spread curve $BS_{CZK\ FX,OIS}$ of EUR/CZK OIS cross-currency swap (EONIA vs CZEONIA + spread).

Table 3. Calculated basis swap spreads $BS_{CZK\ FX,OIS}$, $BS_{EURCZK\ OIS-OIS}$ and quoted $BS_{EURCZK\ FX}$ on 7 November 2015.
Source: Bloomberg, Author's calculations

Maturity	$BS_{CZK\ FX,OIS}$	$BS_{EURCZK\ OIS-OIS}$	$BS_{EURCZK\ FX}$
1 YR	-83.65	5.35	-89.00
2 YR	-77.35	8.65	-86.00
3 YR	-67.10	9.90	-77.00
4 YR	-56.50	10.50	-67.00
5 YR	-46.70	11.30	-58.00

6 YR	-39.95	12.55	-52.50
7 YR	-34.85	13.65	-48.50
8 YR	-32.30	13.20	-45.50
9 YR	-28.00	15.50	-43.50
10 YR	-28.45	13.55	-42.00
12 YR	-13.00	26.50	-39.50
15 YR	-9.10	26.90	-36.00
20 YR	-3.20	28.30	-31.50

In Table 2, we have decomposed EUR/CZK basis spread into EUR/CZK OIS basis spread $BS_{CZK\ FX,OIS}$ and spread that shows the difference of credit and liquidity risks between CZK and EUR rates $BS_{EURCZK,OIS-OIS}$, which is notably higher for Pribor rates than Euribor rates and also has a clear increasing trend with longer maturities. Spread $BS_{CZK\ FX,OIS}$ reflects supply and demand for euros vs Czech koruna and shows higher demand for euros via swap market for all maturities (foreign banks are likely to use FX swap market to offer CZK for EUR pushing CZK yields lower). It is the main factor that contributes to significantly negative basis spread $BS_{EURCZK\ FX}$. In the very long end (above 12 years) the main contributor to negative basis spread is the liquidity and credit risk premium of CZK rates.

The approximation of EUR/CZK OIS cross-currency swap and a practical example on choice of currency for collateral posting concludes this chapter. In the following table we summarize yield curves used to discount future cash flows of derivatives that we discussed in Chapters 3 and 4. These curves depend on the type of derivative and the currency of collateral, for example, CZK FX EUR is the discount curve derived from the collateralized EUR/CZK cross-currency swap, in which we discount the EUR leg with EONIA curve. For simplicity, we limited our cases to CZK, EUR, and USD currencies.

Table 4. Yield curves used to discount future cash flows of collateralized derivatives. The choice of the curve depends on the type of derivative and the currency of collateral.

	IRS CZK	**EUR/CZK BS**		**USD/CZK BS**		**CZK3M CZK6M BS**	
collateral		EUR leg	CZK leg	USD leg	CZK leg	3M leg	6M leg
CZK	CZK OIS	EUR FX CZK	CZK OIS	CZK FX USD	CZK OIS	CZK OIS	CZK OIS
EUR	CZK FX EUR	EUR OIS	CZK FX EUR	USD FX EUR	CZK FX EUR	CZK FX EUR	CZK FX EUR
USD	CZK FX USD	EUR FX USD	CZK FX USD	USD OIS	CZK FX USD	CZK FX USD	CZK FX USD

We have seen that we can approximate the discount curve for the second leg of a cross-currency swap by shifting the swap curve of the first leg. But this curve cannot be used to estimate OIS rates of the second leg because it includes the credit and liquidity spread of the cross-currency swap. We have decomposed basis spread of cross-currency swap into OIS basis spread and a spread reflecting credit and liquidity risk of one currency vs. another one. In the

previous chapter we also discussed the case when there is no market for the two respective currencies and direct quotes against a common currency must be used.

We have focused on CZK market and have shown possible way how to derive CZK OIS curve and have decomposed spread of the EUR/CZK cross-currency spread into individual elements reflecting relative supply and demand and the difference between credit and liquidity risks of CZK and EUR rates. In markets where OIS quotes do not yet exist for all maturities, some assumptions and subjectivity are needed, but the assumptions should be consistent with the requirement of arbitrage-free pricing. Due to increased collateralization, one could expect higher market activity in less liquid OIS markets, including CZK, with more market participants being involved, which could lead to tighter spreads and quotes for longer maturities.

5. Analysis of Cross-Currency Basis Swaps

This chapter discusses determinants that influence cross-currency basis swap spread, in particular, credit and liquidity risks and supply and demand pressures. We show that although an increase in interbank risk in both euro and US dollar caused widening of EUR/USD basis swap spreads, the interbank risk only does not fully capture the level of these spreads. The residual term may be partially assigned to supply and demand imbalances, which may arise and persist over a longer period of time, as is currently the case of EUR/CZK spreads.

The costs associated with temporary swapping one currency into another are expressed as a spread over one leg of the cross-currency swap (CCS). These cross-currency basis spreads are mostly influenced by different credit and liquidity risks of the two underlying money market rates and by supply and demand forces. In this chapter we take a closer look at how these factors influence cross-currency basis swap spreads and discuss arbitrage-free boundaries in cross-currency funding and investing.

As explained in Chapter 3, a floating to floating cross-currency basis swap is a swap that exchanges periodic interest payments based on two money market rates in two different currencies. The exchange rate used to fix the initial and final notional amounts is determined at inception. These are the most commonly used cross-currency swaps and allow users to temporarily transfer assets or liabilities in one currency into another currency. Money market rates (i.e. IBOR rates) in different currencies include different credit and liquidity risk which are reflected as a spread over one leg of the cross-currency basis swap, see Figure 7. The shape of the spread term structure varies over time. The mechanics of currency swaps are well explained e.g. in Baba et al. (2008b). As we discuss below, besides credit and liquidity premium, cross-currency basis swap markets, especially in longer maturities, are also largely driven by demand and supply for different currencies, i.e. by swapped debt issuance.

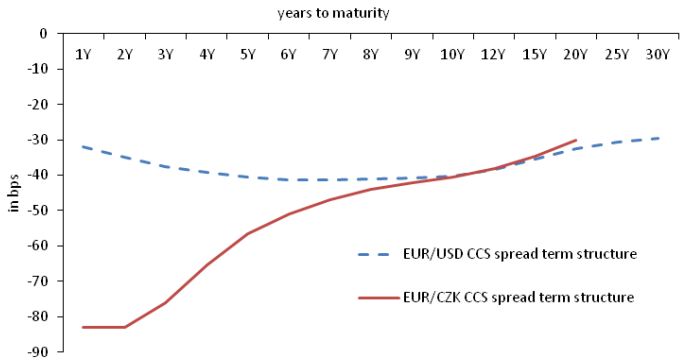


Figure 9. Term structure of CCS spreads of 3M Euribor against 3M USD Libor flat (dotted line) and 3M Pribor against 3M Euribor flat (solid line) as at 5 November 2015.
Source: Bloomberg

CCS basis spreads were traditionally close to zero (apart from bid ask spreads) based on the assumption of banks' continuous access to interbank market financing at IBOR rates. This assumption was widely questioned as spreads have blown in 2007 and practically became an independent market risk factor.

The existing literature discussing cross-currency basis spreads is still somewhat limited. However, factors influencing CCS spreads are often similar to factors influencing single currency swap spreads. There have been a number of papers investigating determinants of interest rate swap spreads, i.e. the differences between government bond yields and interest rate swap levels. For example, Cortes (2006) uses principal component analysis to find that the term structure of swap spreads in different markets move together and are upward sloping in the two to ten year part of the curve due to the existence of a default term premium (a premium reflecting the uncertainty about future interest rates and the risk of systemic failure of the banking sector in the future) and global expectations of government bond issuance (the higher net borrowing, the steeper the yield curve).

Huang et al. (2002) investigated whether liquidity and credit contribute to fluctuations in spreads between 10-year US Treasuries and 10-year USD swaps using vector autoregression method. Their results show that liquidity has a significant negative effect on the swap spreads (swap spreads fall with increased supply and steeper treasury curve). The authors further argue that swap spreads should be mainly regarded as an indicator of market liquidity, and that credit determines the swap spread over longer horizons.

An interesting study was conducted by Heppke-Falk and Hüfner (2004) who analysed if expected budget deficit impact swap spreads in France, Germany and Italy. Using a monthly deficit forecast during the sample period of 1994-2004 they concluded that expected deficit has no significant impact on swap spreads. However, this is contradicted by i.e. Caceres et al. (2010) who show on the recent data that sovereign bond yields rise versus swap yields, when budget deficits increase.

Swap spreads vary considerably over time, and some of this variation has occurred internationally but their term structures also appear to move independently in each market. Also, there might be other specific factors influencing swap spreads such as different liquidity preferences and supply and demand forces.

Ando (2012) presents a study that focuses on EUR/JPY respectively EUR/USD market of cross-currency swaps and investigates reasons for existence of basis swap spreads. Author concludes that the volatility of basis swap spreads is caused by stress in unsecured interbank money

market, especially due to European debt crisis, which cut European banks from unsecured USD financing, but such stress does not explain the whole spread.

Ivashina et al. (2012) present a model in which European banks cut their dollar lending more than euro lending in response to their credit quality deterioration. Banks then raise euro funding through insured retail deposits and swap it to US dollars as they struggle with wholesale dollar funding. Limited demand on the other side makes the synthetic dollar borrowing expensive and banks cut their dollar lending. This model has been successfully tested in the context of recent financial crisis.

Baba et al. (2008b) analyse the spillover effect of the money-market turmoil on FX and CCS markets and show that the use of swap markets to overcome US dollar funding shortages by non-US financials resulted in impaired liquidity in these markets and deviations from covered interest parity conditions.

In following sections we analyse this topic in more detail. We discuss determinants of cross-currency basis swap spreads, and then we decompose these spreads into credit and liquidity elements and test arbitrage-free boundaries via cross-currency funding and investing for two practical cases: EUR/USD and EUR/CZK. Finally, we discuss results and conclude.

5.1. Cross-currency basis spread determinants

Credit, liquidity and supply and demand forces all influence cross-currency basis spreads.

5.1.1. Supply and demand

Cross-currency basis spreads are influenced by the ease of direct access to single currency financing and thus supply and demand for cross-currency financing. Higher amounts of direct borrowing, i.e. government bond issuance impact swap spreads.

The supply and demand balance for temporary swapping two currencies has a larger effect on the long-end of CCS spreads. A *domestic entity* may either

a) fund domestic assets with foreign currency borrowings and use a demand side of the CCS market (a demand for domestic currency) or
b) fund foreign currency assets with domestic currency borrowings and use the supply side of the swap market (a supply of domestic currency).

For example, a) is often used by corporates issuing bonds in foreign currency and swapping proceeds to domestic currency while b) is often used by banks when they lack deposit base in the foreign currency and instead use deposits in their domestic currency.

Both sides are in balance if each of them is able to meet the other side of the trade. A *foreign entity* thus in case of a) issues debt in domestic currency and is a seller of the domestic currency to domestic banks in the CCS market or b) buys domestic assets and is a buyer of a domestic currency from domestic banks in the swap market. If sides of this equation are unequal, it causes volatility and puts pressure on CCS basis spreads.

EUR/USD cross-currency basis swap spreads

The EUR/USD cross-currency swap is used by European banks to fund US dollar assets if other dollar funding sources become inaccessible. The natural other side of this trade are European issuers (agencies, supranationals, and sovereigns) which swap US dollar debt issuance into Euros[11]. European issuers look to issue US dollar bonds and swap the proceeds into Euros in order to diversify their investor base and potentially to obtain cheaper funding. As already discussed, several authors point out (e.g. Ivashina et al. (2012)) that during recent *crisis* period, uncollateralized dollar cash markets were less functional and European banks had to shift to secured transactions such as FX swaps as US money market funds had restrained from buying short term dollar unsecured debt (i.e. CDs, CPs[12]) of European banks. This heavy dependence of European banks on wholesale dollar market during European sovereign debt crisis created a supply and demand imbalance (increased pressure in dollar funding) and EUR/USD cross-currency basis spreads widened.

This, however, goes hand in hand with credit element as widening during periods of volatility reflects the perception of increased credit risk in banks. This was the case recently in the EUR/USD basis swap market, when European banks were perceived by US banks as becoming increasingly riskier, as is empirically investigated in Baba and Packer (2008a). The following graph shows the co-movement of euro-interbank risk (expressed as Euribor-Eonia spread) and EUR/USD basis spreads; an increase in interbank risk causes widening in EUR/USD basis spreads.

[11] Usually the issuer sells US dollar fixed rate bond which is immediately swapped against 3-month USD Libor plus a spread. Then USD Libor payments are swapped against 3-month Euribor payments using cross-currency basis swap so the dollar funding is converted into euro funding. Finally, issuers who use 6-month Euribor as a benchmark enter into basis swap to convert 3-month Euribor payments into 6-month Euribor or a fixed rate. All these steps can be done in a single transaction.

[12] CD – Certificate of Deposit, CP – Commercial Paper

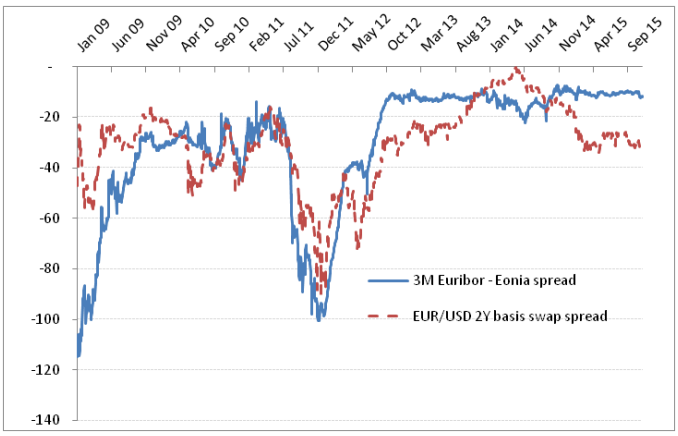


Figure 10. 3M Euribor-Eonia spread in basis points (left hand side, inverted) and EUR/USD 2-year CCS spread since 2009.
Source: Bloomberg

It is important to note that any such explanatory variable varies over time. For example, the above fails to explain the basis spread widening since the second half of 2014. In fact, in June 2014 the ECB announced number of credit expansion steps including cutting cut the deposit rate to negative -0.1%. This second wave of EUR/USD cross-currency basis widening grew stronger with the announcement of ECB's expanded asset purchase programme on January 22, 2015[13]. The ECB has started buying bonds of euro-area governments, agencies, and supranationals within its monthly €60 billion programme that will run until at least September 2016. The initial market impact has been large with yields compressing and curves flattening. Since then basis spreads remained in deep negative territory.

As opposed to 2009, when basis widening was driven by inability of European banks to access unsecured dollar funding, 2015 widening is driven by inability to invest into highly rated EUR denominated government bonds. The general low yield environment in Europe and the negative rate on ECB's deposit facility is pushing investors out of EUR into other currencies like USD. As the ECB purchases have been absorbing large volumes of bonds from secondary markets and driving yields into even more negative territory, investors are looking for currency-hedged investment opportunities abroad.

Central bank actions in terms of supply of currency affect interest rates and borrowing conditions and may cause moves in basis swap spreads. In fact, by simply calculating the ratio of Fed balance sheet to ECB balance sheet, we may construct a simple and rough indicator of relative supply of EUR to USD and compare it to changes in basis swap spread levels (Figure 9).

[13] See ECB press release: http://www.ecb.europa.eu/press/pr/date/2015/html/pr150122_1.en.html

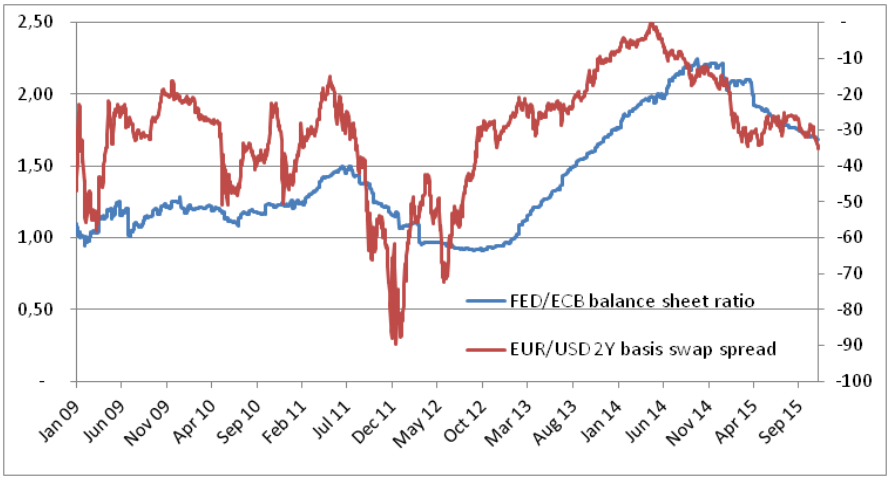


Figure 11. Ratio of Fed to ECB balance sheet (LHS) and EUR/USD 2-year basis swap spread since 2009.

Source: Bloomberg, Author's calculations

We can see from the graph that expansion of Fed balance sheet relative to the ECB balance sheet (Fed balance sheet continued to expand on US Treasury bonds buying while the ECB balance sheet between 2012 and 2014 was shrinking due to repayments of long term refinancing operations) led to basis spread tightening (increased supply of dollars). Since June 2014, it was the ECB's turn to expand while Fed has been decreasing its pace of US Treasuries purchases and halted them in October 2014. This together with the ECB asset purchase programmes created an excess supply of EUR vs. USD and pushed basis spreads wider.

Changes in supply of currency affect changes in market conditions and motivate market participants to borrow or invest in one currency or another. Another supportive driver of basis widening has been the increased EUR issuance by US corporates. Although corporates are not part of the ECB purchase programme, investors looking for a yield pick-up have driven credit spreads of European corporates to significantly tighter levels while USD credit spreads have remained little changed. This makes it attractive for US corporates to tap EUR market.

As already mentioned, agencies, supranationals, and sovereigns are the beneficiaries of wide EUR/USD cross-currency basis as they can obtain cheaper funding in USD. It goes hand in hand that cheaper USD funding for European issuers also means that EUR bonds are more attractive for investors when swapped into USD. Foreign demand for EUR bonds may thus increase from those USD investors, who are able to invest in EUR and swap back into USD. Therefore both USD supply from European issuers and EUR investments from USD investors cause EUR/USD cross-currency basis to tighten. In fact, as we show below, it appears from time to time that for the same issuer, similar bonds in terms of maturities and coupon payments but issued in

different currencies are trading at different credit spreads after fully adjusting for cross-currency basis spreads.

Changes in central banks' balance sheets seem to indicate a general trend of basis spread. However, there is no one single explanatory variable and each of them varies in time. One can see that cross-currency basis swaps do not trade flat and that cross-currency basis spreads are not at or around zero. This holds only theoretically if the two currencies have same credit and liquidity risk, and there is perfect balance between supply and demand.

5.1.2. Short-end of the curve

Although hard to quantify, there are differences between cross currency basis swap drivers in the short end and the long end of the curve. While short end spreads (i.e. in FX swaps) are more influenced by IBOR fixings and credit/liquidity premium, the long end (CCS) is more a function of supply and demand.

Credit element in the short end can be approximated by IBOR-OIS spread[14] while regarding the liquidity premium one can look at the basis spread in cross-currency OIS basis swaps[15]. However, this is only a rough approximation as other factors may play a role, e.g. different IBOR fixing definitions, differences in money market rates and IBOR rates, etc. credit and liquidity premiums were further analysed in Chapter 3.

5.1.3. Long-end of the curve

In the near term, the long maturity CCS swaps have been less volatile than short maturity swaps. Long maturities are largely driven by supply and demand forces in different markets which are further affected by different regulation, market size, liquidity from investors and issuers, etc. Especially very long end might largely depend on debt issuance activity between currencies. Issuance of US dollar bonds from European sovereigns, supranationals and agencies is often swapped backed to EUR and narrows the EUR/USD basis. On the other hand, an increase in swapped euro issuance from US-based corporates widens the basis because demand for USD rises.

To compare difference in credit spreads[16] of bonds with similar cash flows but different currencies we need to adjust them cross-currency basis and interest rate basis, if needed. For example, in case of EUR/USD, we can express credit spread of USD bond in EUR terms as

[14] i.e. the difference between Forward rate agreement (FRA) rate and forward OIS rate with same maturity
[15] i.e. Fed fund vs. Eonia + spread in case of EUR/USD
[16] By credit spreads we mean bond's asset swap spread (ASW)

$$\overline{CS_t^{€}} \approx CS_t^{\$} + BS_t^{€3M-\$3M} - BS_t^{€3M-€6M} \quad (5.1)$$

where $\overline{CS_t^{€}}$ is the synthetic asset swap spread against 6-month Euribor of the US dollar bond with maturity t and asset swap spread of $CS_t^{\$}$, $BS_t^{€3M-\$3M}$ is the EUR/USD cross—currency basis spread for maturity t, and $BS_t^{€3M-€6M}$ is the EUR interest rate basis swap spread (adjusting for interest rate basis is in this case necessary as asset swap spread in USD is marked against 3-month USD Libor while asset swap spread in EUR is by default expressed against 6-month Euribor).

The following graph shows credit spreads of European Investment Bank (EIB) bonds issued in GBP and USD after swapping to EUR and adjusted for 3 to 6-months basis. For short-term maturities, USD bonds appear cheaper than EUR or GBP bonds. For long-term maturities, the differences in credit spreads of individual bonds due to demand and supply imbalances are larger.

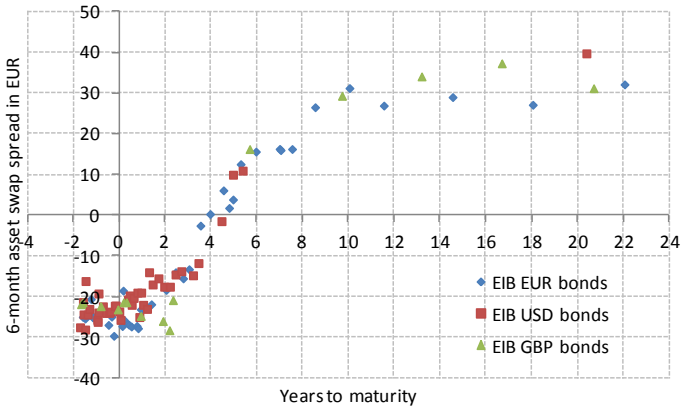


Figure 12. Asset swap spread term structure of EIB bonds in GBP, USD and EUR, i.e. term structure of spreads vs. 6M Euribor in September 2013 . *Source: Bloomberg, Author's calculations*

In the medium to longer-term outlook, the long end of the basis also follows the differences in credits of European and US financials[17], as shown in the following graph. Part of the recent narrowing in the basis can be explained by declining CDS levels for European banks, however, this explanation does not hold anymore because, as discussed above, diverging monetary policies of the ECB and the Fed have had a major impact on basis spread widening since June 2014.

[17] Approximated by Markit iTraxx Europe Senior Financial CDS index for European banks and by S&P/ISDA CDS U.S. Financials Select 10 Index for US banks.

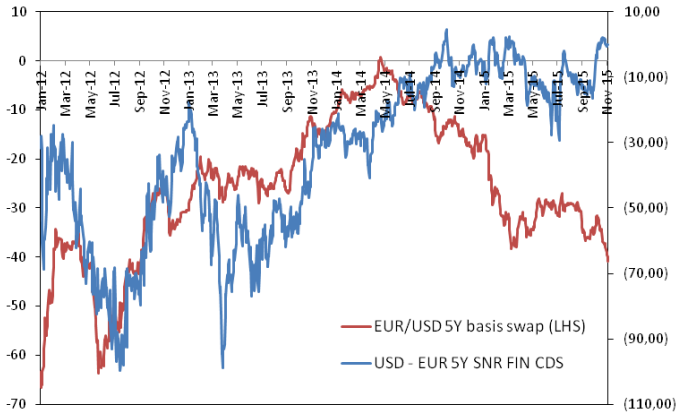


Figure 13. Evolution of 5-year EUR/USD basis spread (LHS) and a spread between 5-year European and US financials CDS (Credit Default Swaps).
Source: Bloomberg

The exact decomposition of basis spreads remains challenging, as cross-currency basis swap spreads reflect both a combination of changes in risk of underlying money market instruments and supply and demand imbalances. Nevertheless, in the following section, we investigate cross-currency basis swap spreads drivers in further detail using multiple regression analysis.

5.1.4. Regression analysis of EUR/USD cross-currency basis swap spreads

We have discussed drivers behind cross-currency basis swap spreads, in particular, credit and liquidity risks, and supply and demand. Our objective in this section is to explain these drivers and their individual importance in determining basis swap spreads with a statistical approach using ordinary least square multiple regression. We test two models, the first with a dependent variable 3-month EUR/USD basis swap spread (to explain short term basis spread determinants), and the second with the dependent variable 5-year EUR/USD basis swap spread (to explain medium-term basis spread determinants). We explain the 3-month and 5-year spread with following independent variables.

To capture short term credit risk for euro and dollar rates, we use IBOR-OIS spreads. For example, if credit risk of European banks increase, we would expect basis swap spread to widen on increased concerns of counterparty credit risk (Figure 10). To capture medium term credit risk of European and US banks, we consider their CDS indices as displayed in Figure 13.

To capture changes in supply of each currency, we investigate Fed's and ECB's balance sheets. As hinted in Figure 11, when the Fed balance sheet expands relative to the ECB balance sheet, the basis spread tends to tighten (increased supply of dollars).

We also add EUR/USD spot rate to see if FX spot market affects CCS basis spreads. We could expect that if the depreciation of Euro against dollar causes forward buying of Euro by corporates, it may cause also widening of the CCS basis spread due to a risen need to hedge these forwards by banks.

Independent variables in regression equation are thus the following

- $€ST\ spread = Euribor_{3M} - Eonia_{3M}$ (in %)
- $\$ST\ spread = USD\ Libor_{3M} - USD\ OIS_{3M}$ (in %)
- $Fed/ECB\ ratio = Fed\ balance\ sheet/ECB\ balance\ sheet$
- $€LT\ spread = EU\ banks\ CDS$ (Markit iTraxx Europe Senior Financial CDS index) (in bps)
- $\$LT\ spread = US\ banks\ CDS$ (S&P/ISDA CDS U.S. Financials Select 10 Index) (in bps)
- $EUR/USD = EUR/USD\ spot\ rate$

To detrend the data we calculate for each variable their weekly changes. Data sample consists of 223 observations between September 2011 and December 2015, and data are taken from Bloomberg (data for US financials CDS index are only available since September 2011). We regress weekly changes in 3-month and 5-year EUR/USD basis swap spreads against weekly changes in the above selected drivers. Our regression equation is

$$\Delta BS_{3M} = \beta_1 \times \Delta EUR/USD + \beta_2 \times \Delta€ST\ spread + \beta_3 \times \Delta€LT\ spread + \beta_4 \times \Delta\$ST\ spread + \beta_5 \times \Delta\$LT\ spread + \beta_6 \Delta \text{Fed/ECB ratio} + \varepsilon$$

$$\Delta BS_{5Y} = \beta_1 \times \Delta EUR/USD + \beta_2 \times \Delta€ST\ spread + \beta_3 \times \Delta€LT\ spread + \beta_4 \times \Delta\$ST\ spread + \beta_5 \times \Delta\$LT\ spread + \beta_6 \Delta \text{Fed/ECB ratio} + \varepsilon \quad (5.2)$$

Regression results are summarized in the following tables.

Table 5. Regression output and summary statistics for 3M (LHS) and 5Y (RHS) changes in EUR/USD basis swap spreads. *Source: Author's Calculations, EViews*

Dependent Variable: D(EURUSD_CCS_3M)
Method: Least Squares
Date: 02/14/16 Time: 19:38
Sample (adjusted): 197 419
Included observations: 223 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(EURUSD)	22.82999	14.19174	1.608682	0.1091
D(EURIBOR3M-EONIA3M)	-85.10708	10.09023	-8.434604	0.0000
D(EUR_FINANCIALS_CDS)	-0.023668	0.026043	-0.908813	0.3645
D(USD_LIBOR3M-USD_OIS...	41.15974	18.63391	2.208863	0.0282
D(US_FINANCIALS_CDS)	-0.209641	0.037842	-5.539918	0.0000
D(FED/ECB)	23.23463	8.936591	2.599943	0.0100

R-squared	0.502362	Mean dependent var	0.386211
Adjusted R-squared	0.490896	S.D. dependent var	4.368832
S.E. of regression	3.117228	Akaike info criterion	5.138303
Sum squared resid	2108.614	Schwarz criterion	5.229976
Log likelihood	-566.9208	Hannan-Quinn criter.	5.175311
Durbin-Watson stat	2.009038		

Dependent Variable: D(EURUSD_CCS_5Y)
Method: Least Squares
Date: 02/14/16 Time: 19:46
Sample (adjusted): 197 419
Included observations: 223 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(EURUSD)	38.69505	10.20517	3.791711	0.0002
D(EURIBOR3M-EONIA3M)	-16.50121	7.255806	-2.274208	0.0239
D(EUR_FINANCIALS_CDS)	-0.051384	0.018727	-2.743847	0.0066
D(USD_LIBOR3M-USD_OIS...	-32.57314	13.39950	-2.430922	0.0159
D(US_FINANCIALS_CDS)	-0.054192	0.027212	-1.991483	0.0477
D(FED/ECB)	-3.601141	6.426234	-0.560381	0.5758

R-squared	0.342467	Mean dependent var	0.042152
Adjusted R-squared	0.327316	S.D. dependent var	2.733051
S.E. of regression	2.241575	Akaike info criterion	4.478772
Sum squared resid	1090.351	Schwarz criterion	4.570444
Log likelihood	-493.3830	Hannan-Quinn criter.	4.515779
Durbin-Watson stat	1.880403		

Selected variables explain roughly 50% of the variance in 3-month EUR/USD basis swap spread and over 30% in 5-year EUR/USD basis swap spread (measured by adjusted R^2).

The following graphs display model results fitted to the actual data of weekly changes of both 3-month and 5-year EUR/USD basis spreads.

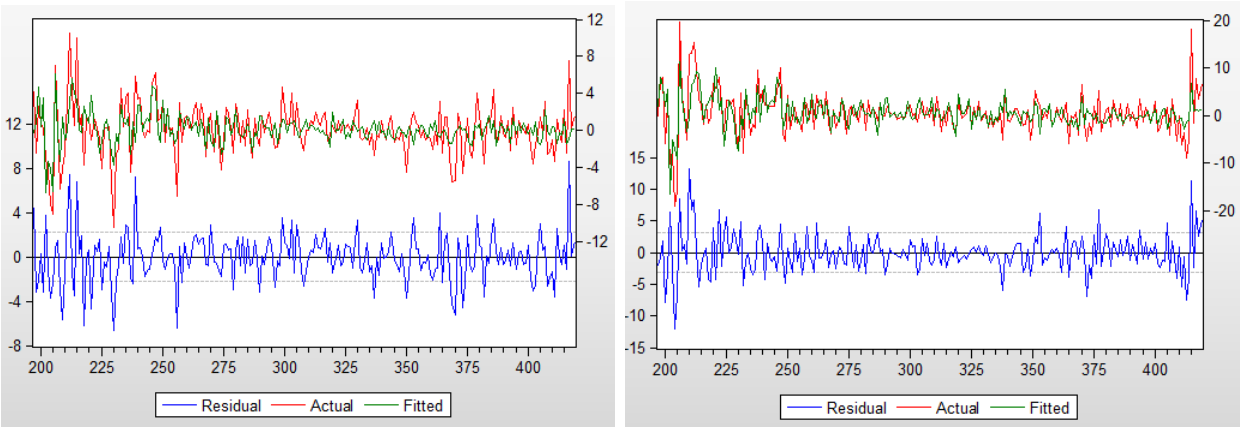


Figure 14. Residuals (left axis), fitted, and actual data (right axis) of 3-month basis spread (LHS) and 5-year basis spread (RHS) *Source: Author's Calculations, EViews*

We can see from the residuals that the model fits better to the middle part of the sample and that fit of the 5-year basis swap seems to have smaller residuals in absolute terms and less outliers. Alsol, the smaller value of each information criteria (Akaike, Schwarz, and Hannan-Quinn Information Criterion) confirms a better model specification for the 5-year basis spread.

3-month EUR/USD basis swap spread

The signs of estimated coefficients confirm our expectations. Negative slope of Euribor3m-Eonia3m confirms that when the credit risk of European banks increase, the 3-month EUR/USD

basis spread widens. Positive slope of Fed/ECB ratio on the other hand confirms that when the Fed balance sheet expands relative to the ECB balance sheet, the 3-month EUR/USD basis spread tightens. We also notice negative coefficient for dollar interbank market stress (USD Libor3m-USD OIS 3m). If credit risk of US banks increase, the 3-month basis spread tightens (one can interpret its regression coefficient β_4 such that 100bps increase in USD Libor-OIS spread would cause 41bps tightening in 3-month basis spread, other variables being held fixed).

Two variables, spot rate EUR/USD and medium term credit risk of European banks, show high p-values and their coefficients are insignificant even at 90% confidence level (last column), and are thus less important drivers of 3-month basis spread.

Lastly, Durbin-Watson statistics very close to 2 does not suggest autocorrelation problem in the residuals.

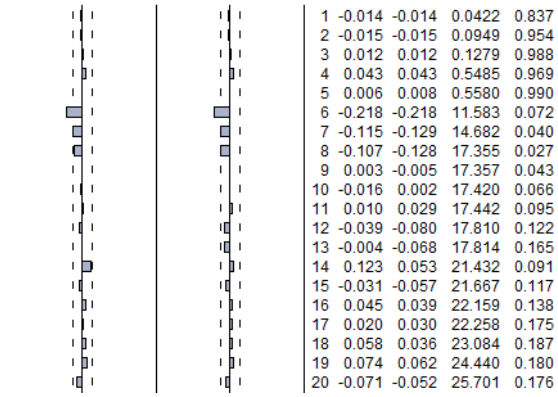
5-year EUR/USD basis swap spread

Interestingly, the exchange rate has a positive effect on 5-year EUR/USD basis spread. If euro appreciates against dollar, 5-year EUR/USD basis spread tightens (becomes less negative). Fed/ECB ratio coefficient is the only insignificant coefficient at 5% confidence level. All other coefficients are significant at 5% confidence level and negative; meaning, short-term or medium term increase in the credit risk of either European or US banks pushes the 5-year basis spread wider. Durbin-Watson statistics suggest some positive autocorrelation of residuals.

The following table displays autocorrelation and partial autocorrelation of the residuals, and Ljung-Box Q-statistics for both 3-month and 5-years EUR/USD basis spread regression models.

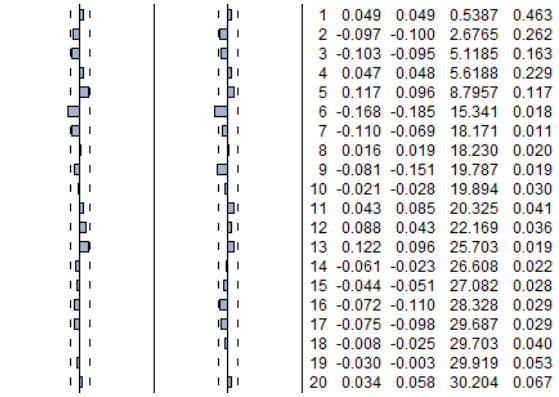
Table 6. Autocorrelation and partial autocorrelation of the residuals for 3-month (LHS) and 5-year (RHS) EUR/USD basis spread regression model. *Source: Author's Calculations, EViews*

Date: 02/14/16 Time: 23:54
Sample: 1 419
Included observations: 223
Q-statistic probabilities adjusted for 6 dynamic regressors



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob*
		1	-0.014	-0.014	0.0422	0.837
		2	-0.015	-0.015	0.0949	0.954
		3	0.012	0.012	0.1279	0.988
		4	0.043	0.043	0.5485	0.969
		5	0.006	0.008	0.5580	0.990
		6	-0.218	-0.218	11.583	0.072
		7	-0.115	-0.129	14.682	0.040
		8	-0.107	-0.128	17.355	0.027
		9	0.003	-0.005	17.357	0.043
		10	-0.016	0.002	17.420	0.066
		11	0.010	0.029	17.442	0.095
		12	-0.039	-0.080	17.810	0.122
		13	-0.004	-0.068	17.814	0.165
		14	0.123	0.053	21.432	0.091
		15	-0.031	-0.057	21.667	0.117
		16	0.045	0.039	22.159	0.138
		17	0.020	0.030	22.258	0.175
		18	0.058	0.036	23.084	0.187
		19	0.074	0.062	24.440	0.180
		20	-0.071	-0.052	25.701	0.176

Date: 02/15/16 Time: 00:08
Sample: 1 419
Included observations: 223
Q-statistic probabilities adjusted for 6 dynamic regressors



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob*
		1	0.049	0.049	0.5387	0.463
		2	-0.097	-0.100	2.6765	0.262
		3	-0.103	-0.095	5.1185	0.163
		4	0.047	0.048	5.6188	0.229
		5	0.117	0.096	8.7957	0.117
		6	-0.168	-0.185	15.341	0.018
		7	-0.110	-0.069	18.171	0.011
		8	0.016	0.019	18.230	0.020
		9	-0.081	-0.151	19.787	0.019
		10	-0.021	-0.028	19.894	0.030
		11	0.043	0.085	20.325	0.041
		12	0.088	0.043	22.169	0.036
		13	0.122	0.096	25.703	0.019
		14	-0.061	-0.023	26.608	0.022
		15	-0.044	-0.051	27.082	0.028
		16	-0.072	-0.110	28.328	0.029
		17	-0.075	-0.098	29.687	0.029
		18	-0.008	-0.025	29.703	0.040
		19	-0.030	-0.003	29.919	0.053
		20	0.034	0.058	30.204	0.067

The correlogram of both autocorrelation and partial autocorrelation suggests that residuals are not serially correlated but a spike can be seen at lags 6 to 8 for 3-month EUR/USD basis spread model with subsequent significant Q-statistics and probabilities. This suggests that model could be further respecified, for example, by using autoregressive model and introducing lags. Similar spike at the 6th lag can be spotted for 5-year EUR/USD basis spread model.

With the above estimation we wanted to illustrate to what extent individual predictor variables explain cross-currency basis swap spreads. This model has several caveats and limitations. The results may largely depend on the studied time period. Furthermore, the number of selected variables could further be minimized and there may be other potential drivers that we have not discussed and tested. One may also question if the selected drivers are exogenous variables or if the basis swap spread may actually determine some of the explanatory variables. Also, explanatory variables and their explanatory power may change over time, thus some kind of a regime-switching or autoregressive model with lags could be further investigated. We aim to answer these questions in the future research.

5.2. Arbitrage-free boundaries for basis swap spreads

In this section we investigate arbitrage-free boundaries for funding and investing in foreign currencies and express FX-implied funding as a function of risk-free rate, interbank market stress and a residual term indicating supply and demand pressure. We discuss two practical cases of EUR/USD and EUR/CZK CCS swap markets.

5.2.1. EUR/USD cross-currency basis swap

We follow Ando (2012) who analysed arbitrage-free boundaries for EUR/USD FX swap implied rates based on funding and risk-free rates. In contrast to Ando (2012), we work directly with EUR/USD basis swap spreads and not with forward and spot FX rates[18]. A EUR/USD basis swap exchanges periodic 3-month USD Libor payments against 3-month Euribor + spread[19]

$$r_{USD\ LIBOR,t} = r_{EURIBOR,t} + BS_t$$

where BS_t is the quoted basis spread for maturity t of the swap contract, and we expect that IBOR funding rates properly reflect the borrowing conditions for domestic banks, and that a

[18] FX swaps follow different mechanics than CCS, however, they both have the same economic function

[19] Although not correct by definition (when calculating present value of future basis swap cash flows), as an approximation, one can subtract/add a spread to one leg or the other:
$r_{USD\ LIBOR,t} = r_{EURIBOR,t} + BS_t$; $r_{USD\ LIBOR,t} - BS_t = r_{EURIBOR,t}$

bank can invest at OIS (overnight indexed swap) risk-free rate in either Euros or US dollars. Natural boundaries for risk-free investing thus become

$$r_{EONIA,t} - r_{EURIBOR,t} \leq BS_t \leq r_{USD\ LIBOR,t} - r_{USD\ OIS,t}, \tag{5.3}$$

If (5.3) does not hold, then arbitrage opportunity arises

- If $r_{USD\ LIBOR,t} - BS_t < r_{USD\ OIS,t}$, then a bank which can access unsecured EUR market will borrow at Euribor, swap the proceeds into USD and invest at risk-free USD OIS rate[20]
- If $r_{EURIBOR,t} + BS_t < r_{EONIA,t}$, then a bank which can access unsecured USD market will borrow at USD LIBOR, swap the proceeds into EUR via FX or cross-currency swap and invest at risk-free EONIA rate.

Further, setting $BS_t = r_{EONIA,t} - r_{EURIBOR,t} + X$, we can use X as an indicator of arbitrage opportunity (for $X < 0$ arbitrage exists). It is now easy to express FX swap implied USD funding rate $r_{FX,t}$ from EUR funding rate (Euribor) in terms of X, IBOR and OIS rates

$$\begin{aligned} r_{FX,t} &= r_{USD\ LIBOR,t} - BS_t = r_{USD\ LIBOR,t} - \left(r_{EONIA,t} - r_{EURIBOR,t}\right) - X \\ r_{FX,t} &= r_{USD\ OIS,t} + \left(r_{USD\ LIBOR,t} - r_{USD\ OIS,t}\right) + \left(r_{EURIBOR,t} - r_{EONIA,t}\right) - X. \end{aligned} \tag{5.4}$$

FX swap implied rate is a function of a forecast of Federal funds rate, stress in the USD and EUR money markets (IBOR-OIS spreads) and a residual term X which partially indicates supply and demand pressure of one currency vs. another. Ando (2012) further explains that $X < 0$ may arise from low liquidity in unsecured USD money markes, specific counterparty risk, transaction costs and measurement errors in the LIBOR fixing rate.

Ando (2012) uses forward and spot prices to determine USD FX-swap implied rates from both EUR and JPY during the period of increased market stress in 2011 and using decomposition similar to (5.4) shows how each factor contributed to widening/tightening of FX basis. The author concludes that when USD supply and demand situation tightens, stress in the unsecured money markets increases and basis spreads widen and that FX-implied rate sometimes reaches levels not explained by the stress in the unsecured markets ($X < 0$).

We can rewrite (5.3) in terms of $r_{FX,t}$ such that $r_{FX,t}$ is bounded by the risk free USD rate from below and by the sum of unsecured USD money market rate and EUR market stress indicator from above

$$r_{USD\ OIS,t} \leq r_{FX,t} \leq r_{USD\ LIBOR,t} + \left(r_{EURIBOR,t} - r_{EONIA,t}\right), \tag{5.5}$$

[20] Or any other proxy of a risk-free rate, i.e. a repo rate with high quality collateral.

If (5.5) holds, then the level of FX-implied dollar rate $r_{FX,t}$ is determined by supply and demand forces which however do not yet create any arbitrage opportunity in our setting.

Applying (5.4) to a 3-month maturity EUR/USD basis swap, one can see that arbitrage opportunity arose and persisted over a significant period of time. The following graph shows the evolution of 3-month FX-implied USD rate from Euribor decomposed into USD risk-free rate and euro and US market stress indicators.

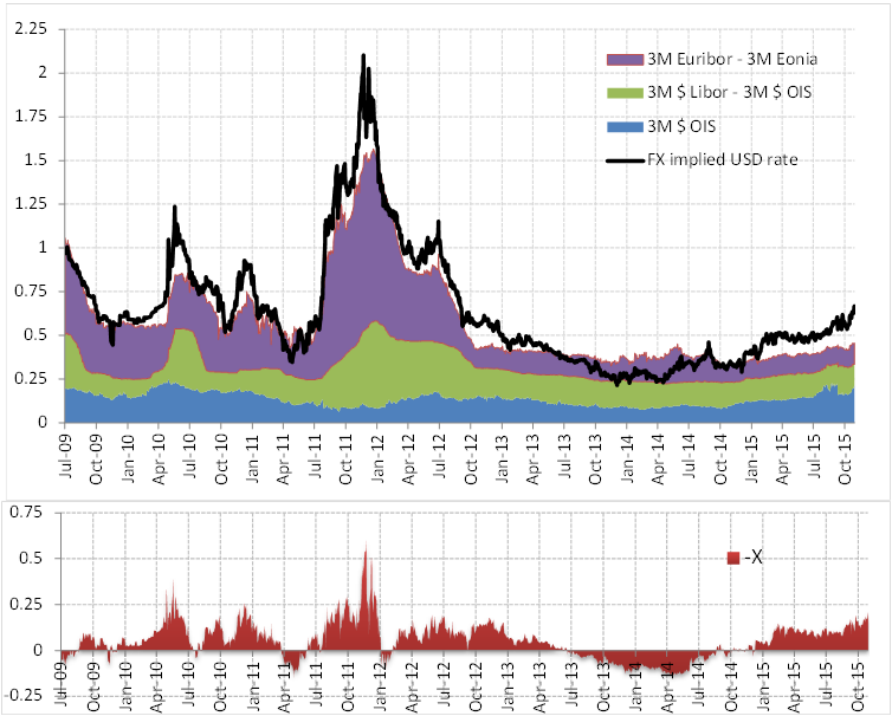


Figure 15. FX-implied USD rate from Euribor decomposed into EUR and USD interbank risk, USD risk-free rate and a residual term indicating supply and demand imbalances.
Source: Bloomberg, Author's calculations

The graph above shows that an increase in interbank risk in both euro area and US (IBOR-OIS widening) causes widening of EUR/USD basis spreads. The interbank risk, however, does not fully capture the movements in basis spread and the residual term indicates supply and demand imbalances. Short-term US dollar issuance had become more attractive for issuers having well-established US market access. An arbitrage opportunity for issuers who were able to raise US dollars at around USD Libor rate arose and persisted over a significant period of time. This suggests that there was only limited capital available to fully exploit this imbalance.

The arbitrage opportunity peaked at the end of 2011 right before coordinated central banks actions that effectively capped the basis spreads. The central banks intervened with cross-currency swap lines, and European banks could borrow dollars directly from ECB against Euro

collateral. Baba and Packer (2008a) present evidence that dollar term funding auctions by the ECB have stabilized the FX swap market. The EUR/USD basis has narrowed significantly during first half of 2014 and arbitrage opportunity vanished. Tighter basis spreads make funding in Euros more attractive for US dollar issuers, and less appealing for Euro issuers to issue in US dollars[21].

Another arbitrage opportunity arose in January 2015 when the ECB announced its public sector purchase programme, which has persisted since then, what suggests that excessive supply of EUR has not yet been absorbed.

5.2.2. EUR/CZK FX swap

The arbitrage case of EUR/CZK FX financing is even more remarkable. We do not have a full history of Czech koruna risk-free rates expressed in terms of CZK OIS rates so we use Czech National Bank's (CNB) main monetary policy instrument, the 2-week repo tender, as the risk-free rate[22]. That is also why we need to use FX swaps instead of CCS. To calculate the FX-swap implied EUR funding from CZK funding rate (Pribor) $r_{FX\ EUR,2W}$, we rewrite $r_{FX\ EUR,2W}$ in terms of quoted EUR/CZK spot rate *S*, 2-week swap points *sp* and Pribor rate $r_{PRIBOR,2W}$

$$S + sp = S\frac{1 + r_{PRIBOR,2W}\frac{D_t}{360}}{1 + r_{FX\ EUR,2W}\frac{D_t}{360}}$$

$$r_{FX\ EUR,2W} = \left[\frac{S}{S+sp}\left(1 + r_{PRIBOR,2W}\frac{D_t}{360}\right) - 1\right]\frac{360}{D_t} \quad (5.6)$$

where D_t is number of days of the contract (14 days, or 2-weeks). We now rewrite (5.4) for the case of EUR/CZK

$$r_{EUR\ FX,2W} = r_{EONIA,2W} + \left(r_{EURIBOR,2W} - r_{EONIA,2W}\right) + \left(r_{PRIBOR,2W} - r_{CNB\ REPO\ 2W}\right) - X, \quad (5.7)$$

where $r_{CNB\ REPO\ 2W}$ is the CNB limit repo rate. FX swap implied EUR rate is a function of a 2-week EONIA swap rate, stress in the EUR and CZK money markets and a residual term X. We rewrite (5.5) for the case of EUR/CZK

$$r_{EONIA,2W} \leq r_{EUR\ FX,2W} \leq r_{EURIBOR,2W} + \left(r_{PRIBOR,2W} - r_{CNB\ REPO\ 2W}\right) \quad (5.8)$$

[21] E.g. on 6 November 2013 the World Bank (IBRD) took advantage of the FX basis swap movements and issued a 3-year €-denominated 1bn benchmark bond at Mid Swaps minus 28bps, a level comparable to their USD levels after swapping. It was their first €-benchmark since 2009. In fact, since then there have been several opportunities for USD investors to replace IBRD 2016 $-bond with IBRD 2016 €-bond swapped into $ for an attractive pick up in yield and spread.

[22] CNB accepts surplus liquidity from banks against CNB collateral with duration of 14 days and pays CNB 2-week repo rate. For the purpose of the calculation, we also assume that bids are satisfied at the limit repo rate.

so that $r_{EUR\ FX,2W}$ is bounded by EUR risk free rate from below and by the sum of unsecured EUR money market rate and Czech interbank risk. Inequality (5.8) must hold not to allow for riskless profit.

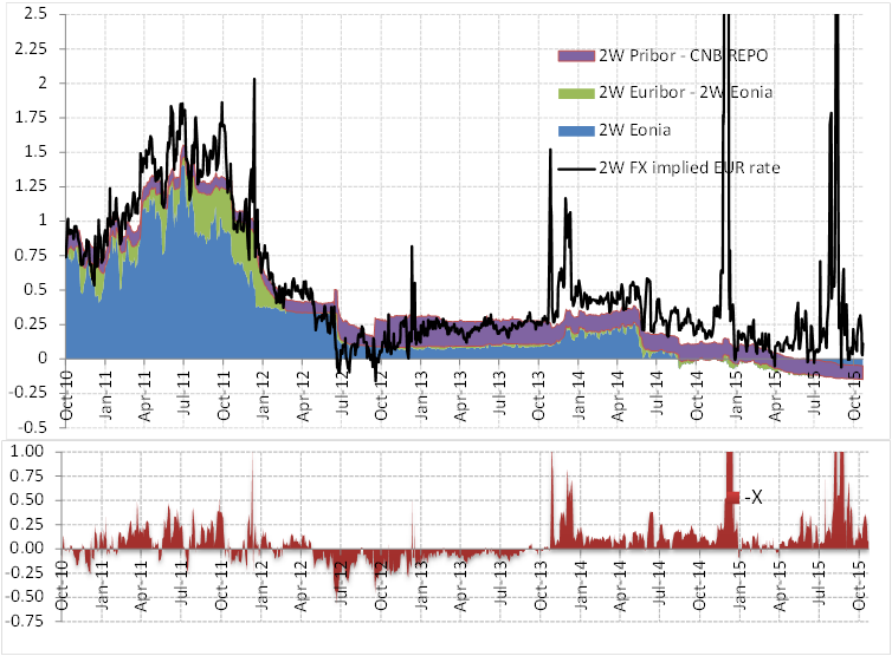


Figure 16. FX-implied EUR rate from Pribor decomposed into EUR and CZK interbank risk, EUR risk-free rate and a residual term indicating supply and demand imbalances.
Source: Bloomberg, Author's calculations

Figure 16 shows that for example, such arbitrage opportunity arose and has since persisted after CNB decided to fight deflation through FX channel. To further ease monetary conditions, CNB started intervening in the FX market and weakening CZK in order to spur inflation and economic recovery[23]. As a result, CZK depreciated, and increased money supply in CZK destroyed short term FX-implied CZK rates pushing them into deep negative territory. For example, a bank that can borrow funds at 2-week Euribor rate and has access to CNB policy instruments can temporarily transfer euros into CZK via FX swap and pay negative yields for such CZK liquidity, which can be deposited at positive 0.05% p.a. at CNB (Figure 17). This arbitrage opportunity may gradually vanish when Czech financial sector is able to absorb excess CZK liquidity from foreign banks that bought CZK from CNB, but currently negative yields on Czech government bills and short term bonds, together with implicit CZK negative rates from EUR/CZK FX swaps suggest that this has not yet been the case and foreign banks still hold excessive CZK amounts.

[23] On 7 November 2013 CNB Bank Board committed to keep the exchange rate close to CZK 27 to EUR until the inflation target is reached and there are concerns about price stability (rise in inflation).

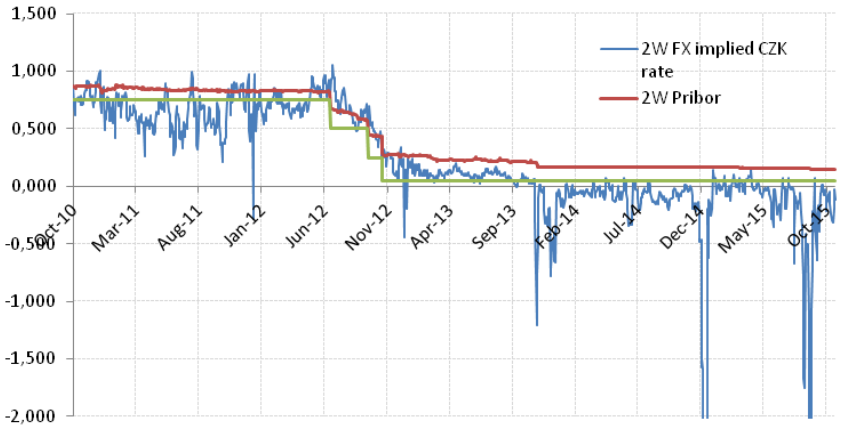


Figure 17. 2-week FX-implied CZK rate from Euribor, 2-week Pribor rate and CNB repo rate.
Source: Bloomberg

Similar analysis can be easily replicated choosing different maturity (e.g. 3-months) and different CZK risk-free rate (e.g. 3-month government T-bill). However, in that case, one could rather focus on specific instrument and maturity due to historical availability of data. In practice, one has to account for bid-offer spreads and as we have seen IBOR rates are different to real borrowing rates of individual banks, however, there is no loss of generality in equations (5.6)-(5.8) and they can be easily applied to other cases and tenors.

To conclude, this chapter discussed factors that influence cross-currency basis swap spreads, in particular, credit and liquidity risks and supply and demand pressures. We have discussed that basis spreads in the short-end of the curve are more influenced by IBOR fixing and credit/liquidity premium while the long-end is more a function of supply and demand. We built regression models for changes in 3-month and 5-year EUR/USD basis swap spread that explained 50% and 30% of the variance of weekly spread changes. The model confirmed, for example, that increased credit risk in European banks widens the EUR/USD basis spread. But interbank risk only does not fully capture the level of basis spreads. The residual term may be partially assigned to supply and demand imbalances that may arise and persist over a longer period of time. In the case of EUR/CZK, we have shown that recent FX-intervention distorted Czech money market that has not yet been able to absorb excessive CZK money supply.

6. Valuation of Zero-coupon Inflation Swaps

The market for inflation linked products has grown significantly in recent years. Government issuers have been the main driver behind the growth of inflation-linked (I/L) bond markets. I/L bonds offer investor protection of purchasing power as their coupons and principal payments are linked to changes in price indices. Inflation derivatives have been developed together with I/L bonds and are used to manage inflation risk.

In this chapter we describe main elements of zero-coupon inflation linked swaps including their use, their mechanics, and their pricing. The aim of this chapter is similar; to present a consistent valuation approach to zero-coupon inflation swaps that properly reflects the risk of its cash-flows however we provide a general overview on functioning of inflation markets as well. We first provide a brief review of relevant studies on inflation market. Then we offer an insight into the inflation market, its main participants, inflation measurement, and breakeven inflation rates. We also discuss the real yield of inflation-linked bond ("linkers") versus the nominal yield of the traditional bond. Then we touch on mechanics of zero-coupon inflation swap and present a consistent approach to its valuation. This approach is further illustrated using OIS discounting for the case of zero-coupon inflation swap collateralized with cash.

From a number of studies that analyse inflation markets we refer to the ones that focus on market liquidity of inflation swaps, price discovery of breakeven inflation, and relationship between nominal yields, real yields, or inflation expectations.

Fleming and Sporn (2013) study trading activity and pricing transparency of zero-coupon inflation swap trades in the US. They find that despite low trading volumes and number of trades and their OTC nature, the market is reasonably transparent and liquid, in particular, the traded prices of the sample were close to end-of-day quoted prices and that customer buy and sell prices essentially matched bid-ask spreads quoted by the dealers.

Schulz and Stapf (2011) analyse price discovery for breakeven inflation rate using high frequency data in the US and the euro area and concluded that during crisis price formation was concentrated mostly in government bonds (due to their liquidity and low credit risk). In general, for maturities up to 5 years, price discovery comes from both swap and bond market, for longer maturities bond markets dominate.

Haubrich et al. (2012) introduce a model to analyse term structure of nominal, real, and inflation rates from inflation swaps, government bonds, and government inflation linked bonds

(TIPS) in the US and found that TIPS were underpriced in their early years and then during the period of recent financial crisis.

Bauer (2014) shows that changes in inflation expectations alone or when caused by macro data surprises explain a sizable share of changes in nominal rates.

Regarding pricing of inflation linked derivatives, Mercurio (2005) derives analytical formulas for inflation indexed swaps and caps using FX analogy and looking at real rates as nominal rates in foreign economy and at CPI as an exchange rate between nominal and real rates.

On the other hand, there are not many recent papers focusing on valuation of inflation derivatives that take into account the risk of the derivative, which affects the choice of discount rates. In Section 2, we present such valuation approach for zero-coupon inflation swap and discuss the proper choice for discounting curves based on the risk of the trade and the way this risk is mitigated. We then illustrate this on the example of using OIS discounting for zero-coupon inflation swap collateralized with cash.

6.1. Inflation Market

The inflation bond market started in the UK with the first inflation-linked (I/L) Gilt in 1981. However, the global market started to develop after the US Treasury began issuing TIPS (Treasury Inflation Protected Securities) in 1997. France issued the first bond linked to French inflation in 1998. In 2001, France started issuing government bonds linked to European inflation (Euro HICP), and took advantage of a liquid market with wider investor base. Italy, Greece, and Germany soon followed and many other sovereigns have since been able to develop inflation-linked securities in their local markets. Most of the issuance has been however concentrated in the US, UK and France.

Main issuers of I/L securities are governments and government agencies. Governments have partly natural needs to issue I/L bonds since tax revenues are linked to inflation. Issuing I/L bonds is also convenient for governments as it allows to increase the duration of their debt portfolio.

The growth and turnover of the I/L market has increased supported by overall lower levels of inflation and inflation targeting as the primary monetary policy objective, and availability of statistical inflation data. In the emerging markets, growth expectations also increased the demand for inflation-linked bonds.

Inflation derivatives have been developed together with I/L bonds. In general, market participants use inflation derivatives or I/L bonds to manage their assets and liabilities for inflation exposure. Inflation products are thus used to transfer inflation risk of assets or liabilities. Investors also buy I/L bonds to diversify their investments and issuers issue I/L bonds to diversify their funding sources (and expand their investor base), or to exploit relative value opportunities.

As previously mentioned, governments are the main I/L issuers and inflation payers. From private sector, inflation sellers include companies with revenues linked to infrastructure projects. For example, real estate investors may hedge inflation risk of rental revenues where rents are indexed to inflation. On the other side, receiving inflation protects investors' real return. Pension funds and insurance companies use inflation derivatives or I/L bonds to hedge those pensions that are linked to inflation. Banks are buyers of inflation-linked government bonds in general when they asset swap it against LIBOR rates that measure their short term funding costs. Further, banks may want to receive inflation to hedge their I/L retail products. Asset managers buy inflation for those funds that are inflation-benchmarked.

6.1.1. Measuring Inflation

Consumer Price Index (CPI) is used as a standard measure of inflation. It is defined and calculated by statistical agencies, and it measures the cost of a basket of consumer goods and services in nominal prices. It is often calculated once a month and published around two weeks later.

The inflation rate between time T_0 and *T* is measured as the change in the CPI index

$$\frac{Index_T}{Index_{T_0}} - 1,$$

where $Index_T$ is the price of the basket at time *T*, and $Index_{T_0}$ is the price of the basket at time T_0. Annual inflation rate is considered as the most common reference for inflation. It is measured by the change in the CPI over 12 months

$$\frac{Index_T}{Index_{T-12}} - 1.$$

Euro HICP Index

Different inflation indices in different countries may give distorted picture about inflation rates among countries. To measure inflation consistently, same methodology or index definition should be applied, such as Harmonized Index of Consumer Prices (HICP), which makes it possible to compare rates of inflation across European countries. Eurostat calculates the Euro-area inflation rate as a weighted average of national indices (excluding tobacco) of all countries in the Eurozone and publishes Euro HICP flash estimate usually on the last working day of the reference month and revised data approximately 18 days later (see Eurostat (2015)). Euro HICP Index is used as the reference index for European I/L bonds and the European Central Bank uses HICPs as the reference measure of inflation in the Euro-area countries.

6.1.2. Inflation-Linked Bond

The purpose of a bond linked to inflation index is to preserve the value of money and the purchasing power for investor. It guarantees an investor a real return as both the notional of the bond and coupons rise with inflation. In year *T*, the notional and the coupon equal to

$$N_T = N_{T_0} \times \frac{Index_T}{Index_{T_0}}, \text{ where } N_{T_0} \text{ is the original bond notional.}$$

$$C_T = C_{T_0} \times \frac{Index_T}{Index_{T_0}}, \text{ where } C_{T_0} \text{ is the real coupon.}$$

The real coupon is usually set lower than a coupon of a conventional bond if positive inflation is expected. In this case, the inflation-linked bond pays lower coupons during bond's life but the redemption value at maturity is higher (adjusted for positive inflation). Therefore, the duration of I/L bond is also longer. However, the duration of a linker represents the relationship of a price change to change in real yield, while the duration of the traditional bond takes the nominal yield into account. Therefore, the duration of a linker cannot be directly compared to the duration of the traditional bond.

The yield of a linker is quoted on a real basis; it is calculated from real coupons and notional without adjusting for inflation expectations. To make the real yields of a linker comparable with nominal yield of a traditional bond, inflation expectations must be added to the real yield.

In case of a zero-coupon inflation-linked bond, which has a single payment I_T at maturity *T*, the nominal present value DF_T^{IL} of such bonds includes inflation expectation and is different from nominal present value of standard zero-coupon bond DF_T^N

$$DF_T^{IL} = DF_T^N I_T. \tag{6.1}$$

To calculate the real present value of I/L bond DF_T^R (its market value), the nominal present value of I/L bond DF_T^{IL} is divided by the current inflation index I_0

$$DF_T^R = \frac{DF_T^{IL}}{I_0} = \frac{DF_T^N I_T}{I_0}. \tag{6.2}$$

The nominal yield on I/L bond r_T^{IL} can be then derived as

$$r_T^{IL} = \left(\frac{I_T}{I_0 DF_T^R}\right)^{\frac{1}{T}} - 1. \tag{6.3}$$

Because the payment at maturity I_T is not known, the nominal yield on I/L bond is also not known, but the real yield r_T^R is locked in and can be calculated as

$$r_T^R = \frac{1}{\left(DF_T^R\right)^{\frac{1}{T}}} - 1. \tag{6.4}$$

Inflation-linked bonds in the Czech Republic

The Czech Republic has issued three retail inflation-linked bonds in recent years, and subsequently tapped two of them (see Table 5 for an overview). The bonds are puttable with put options exercisable every 6 months.

Table 7. Characteristics of Czech retail inflation-linked bonds.
Source: http://www.sporicidluhopisycr.cz/

ISIN	CZ0001003586	CZ0001003586 (tap)	CZ0001003990	CZ0001004204	CZ0001004204 (tap)
Subscription	10. 5. – 31. 5.	5. 11. – 30. 11.	13. 5. – 31. 5.	4. 11. – 29. 11.	19. 5. – 4. 6.
Issue date	12.6.2012	12.12.2012	12.6.2013	12.12.2013	12.6.2014
Maturity	12.6.2019	12.6.2019	12.6.2020	12.12.2020	12.12.2020
Denomination	1 CZK	1 CZK	1 CZK	1 CZK	1 CZK
Issue price	100%	100%	100%	100%	100%
Coupon rate	% change in CPI index	% change in CPI index	% change in CPI index + 0.25%	% change in CPI index + 0.25%	% change in CPI index + 0.25%
Coupon frequency	6 months	6 months	6 months	6 months	6 months
Amount Issued	1 270 385 360 CZK	834 418 829 CZK	411 238 606 CZK	1 369 405 301 CZK	663 204 577 CZK

The yield of inflation-linked bond is different from inflation rate published by Czech Statistical Office (CZSO). The difference stems from using different basic indices and methodology. For example, the average annual inflation rate published by the CZSO shows the ratio of the sum of

the last 12 months CPI values over the sum of the previous 12 months (12 month moving average). Year-on-year (yoy) inflation rate is the yearly percentage change in CPI levels calculated as a ratio of last month's CPI level to the CPI level of the corresponding month in the previous year while month-on-month (mom) inflation rate compares the change in last month's CPI level to the preceding month, see CZSO (2015).

Both measures are however different from the yield of inflation-linked bond. The bond pays a six month coupon, however, the coupon rate is not calculated as the average 6-month inflation rate, but rather as the percentage change in CPI level of the current month and the CPI level six months back. The current month is lagged by two months from the month of the coupon payment date. The coupon is further floored at zero to protect investor from deflation. Because the coupon is not paid but reinvested, the coupon rate is higher or equal than the yoy inflation rate (higher in case of deflation) and the buyer is thus protected from inflation.

6.1.3. Breakeven inflation rate

The breakeven inflation rate is used to compare yields of inflation-linked and traditional bonds; it is the average annual inflation rate that needs to be realized to make the yields of the traditional bond and the inflation-linked bond equal. It is calculated by making the present value of cash flows of the linker discounted with the yield of the traditional bond equal to the price of the linker. If the breakeven inflation rate is lower than expected inflation rate, then investor would buy the inflation-linked bond.

To assess breakeven inflation rate, it is also possible to use a simpler Fisher equation that shows the relationship between nominal yield r_T^N and real yield r_T^R

$$(1 + r_T^N) = (1 + r_T^R) \times (1 + b)$$
$$b = \frac{(1+r_T^N)}{(1+ r_T^R)} - 1, \tag{6.5}$$

or its first approximation

$$r_T^N \approx r_T^R + b.$$

The breakeven inflation b rate can be thus approximated as the difference between the nominal yield of the traditional bond r_T^N and the real yield of inflation-linked bond r_T^R. If we ignore credit and liquidity risk premium of the inflation-linked bond, and the compounding effect linked to inflation-linked bonds, then the breakeven inflation is the market expected inflation over I/L bond's life.

6.2. Zero-coupon inflation swaps

An inflation swap is a contract in which one party exchanges regular fixed or floating interest rate cash flows in return for floating rate determined by an inflation index. Inflation swaps are used to hedge inflation risk or speculate on future levels of inflation as they reflect market expectations of future inflation rates.

The basic inflation swap is a zero-coupon inflation swap, in which a buyer (inflation receiver) pays a fixed rate payment to the seller (inflation payer) in return for inflation-linked payment at maturity date. There is no exchange of cash flows at inception. Therefore, it is practically only an exchange of one net cash flow on the maturity date and the maturity date is a number of years after the start date.

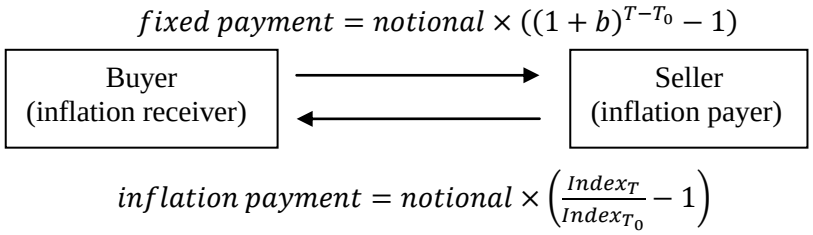


Figure 18. Cash flows at maturity of zero-coupon inflation swap

In this section we base the valuation approach on Kerkhof (2005) and develop it further to take into account the risk of swap cash flows. We also mention the case of OIS discounting, that is, when the transaction is fully collateralized with cash with interest payments linked to OIS rates.

6.2.1. Valuation of zero-coupon inflation swap

In the zero-coupon (ZC) inflation swap the parties specify the base month for inflation index and the end month. Inflation buyer pays the agreed compounded fixed rate and receives the change in inflation index for the inflation period. The inflation period is given by the base month and the end month. If the inflation swap is spot starting, then the value of the inflation index for the base month is known.

The fixings of inflation indices for inflation bonds or derivatives are lagged usually by two or three months. The inflation payment is calculated either by interpolating between lagged monthly index levels, where the linear interpolation is given by the trade date (*interpolated index*), or by using *monthly index* levels lagged by two or three months for any trade date during month. The convention and the number of month lags depend on the market. For

example, if the inflation index is the European HICP index with base month being three months before the trade date, then the value of such index for the base month is published around two months before the trade date, and is therefore known at the trade date. At maturity, value of the index for the end month would be known as well). If the index value for the base month is not known, then it is forward starting inflation swap.

For spot starting ZC inflation swap, the index $Index_{T_0} = I_{T_0}$ for the base month is known while the index for the end month $Index_T = I_T$ is the only unknown. The breakeven inflation rate b is the quoted fixed rate in the market.

As for any interest rate swap, at inception, the present value of the fixed leg must equal the present value of the inflation leg

$$PV(fixed\ leg) = PV(inflation\ leg),$$

The present value of the fixed leg is the discounted compounded fixed rate

$$PV(fixed\ leg) = DF_T^N((1+b)^{T-T_0} - 1). \tag{6.6}$$

The present value of the inflation leg is the value of the change in inflation index for the period $T - T_0$. From (6.2) we have $DF_T^{IL} = I_0 DF_T^R$; therefore

$$PV(inflation\ leg) = DF_T^N\left(\frac{I_T}{I_{T_0}} - 1\right) = \frac{I_0 DF_T^R}{I_{T_0}} - DF_T^N, \tag{6.7}$$

where DF_T^N is the nominal discount factor. The present value of both legs thus depends on discount factors DF_T^N. When calculating DF_T^N, proper discounting method needs to be chosen to take into account the risk of the cash flows. The traditional approach used swap rates (with floating rates linked to LIBOR) to construct the discount curve, but this approach is not arbitrage free given the presence of credit and liquidity risks. The value of the transaction depends on the risk of the transaction and the way this risk is mitigated during the life of the trade. The traditional approach has been further developed by the market to take into account the risk of default of the counterparty, the risk of one's own default, or the interest paid on the collateral as discussed in previous chapters.

To illustrate how discounting affects the valuation of zero-coupon inflation swap, we consider the case of cash collateralized swap (we assume that the swap is fully collateralized with cash which pays the interest rate equal to overnight index average rate and that the OIS market exists for all maturities). Then DF_T^N is bootstrapped from the overnight indexed swap (OIS) rates as in (3.11)

$$DF_T^N = \frac{1 - r_T \sum_{t=1}^{T-1} \delta_t DF_t^N}{1 + \delta_T r_T}, \tag{6.8}$$

where r_T is the OIS swap rate for period T and δ_t is the year fraction between *t-1* and *t*. As discussed in previous chapters, OIS discounting has recently become a market standard for valuation of cash collateralized derivatives.

The only unknown in (6.7) thus remains the real discount factor DF_T^R. Because PVs of both swap legs are equal, DF_T^R can be easily derived as

$$DF_T^R = \frac{I_{T_0}}{I_0} DF_T^N (1+b)^{T-T_0}. \tag{6.9}$$

6.2.2. Construction of zero coupon inflation swap curve

To value existing swap we need an inflation curve, i.e. the breakeven reference rates not only for years, but for all months in between years. An inflation curve is constructed from market prices of zero-coupon inflation swaps combined with seasonality adjustments. Quotes of annual zero-coupon inflation swaps provide market projections of CPI increase over the term of the swap (projected future CPI fixings). From projected CPI fixings, one can interpolate between two values to calculate annual trend in inflation. A linear, cubic or other interpolation method can be used. Because of seasonal behaviour of CPI indices it is not possible to assume that CPI indices grow constantly; therefore the interpolation has to be finally adjusted for seasonality.

Projected CPI index for the index reference date *T* years after starting reference date ($PR\ Index_T$) is calculated as

$$PR\ Index_T = Index_{T_0}(1+b_T)^T, \tag{6.10}$$

where b_T is the quoted zero-coupon swap fixed rate with maturity *T* years (*T* years breakeven rate) and $Index_{T_0}$ is CPI index value for the start date. Choosing linear interpolation between $PR\ Index_T$ and $PR\ Index_{T+1}$ we obtain the annual inflation rate trend (growth rate in breakeven rate) as

$$annual\ trend = \ln\left(\frac{PR\ Index_{T+1}}{PR\ Index_T}\right). \tag{6.11}$$

A common approach is to use continuously compounded forward inflation rate. The annual trend in fact determines yearly forward inflation rates unadjusted for seasonality. Projected CPI for date *T* years and *m* months after the start date and before seasonal adjustment can be then calculated as

$$PR\ Index_{T+m} = Index_{T_0} e^{\ln\left(\frac{PR\ Index_{T+1}}{PR\ Index_T}\right)\frac{m}{12}}. \tag{6.12}$$

Finally, we need to incorporate monthly seasonality to get seasonally adjusted projected CPIs values by adding a seasonal component to annual trend

$$PR\ Index_{T+m} = Index_{T_0} e^{\ln\left(\frac{PR\ Index_{T+1}}{PR\ Index_T}\right)\frac{m}{12}} e^{\sum_{i=1}^{m} sc_i}, \tag{6.13}$$

where sc_i is monthly seasonal component for *i-th* month after start month[24]. The sum of seasonal adjustment components should equal to zero so they do not affect quoted swap rates. Further, it is often assumed that seasonality is a set of 12 monthly seasonal components that remain constant over time.

Seasonality of inflation

Seasonality and volatility of inflation can largely affect the inflation trend and often depends on commodity prices (food and energy prices) or changes in fiscal policy (i.e. tax changes). Seasonality in prices of inflation-linked bonds and swaps is in general not accounted for when looking at breakeven rates (inflation products are linked to price indices that are unadjusted for seasonality). Seasonal adjustment is therefore necessary to uncover the trend in inflation rates outside standard yearly maturity dates. Seasonality can be measured either by statistical methods (for example by using autoregressive moving average models, typically X-12 ARIMA[25]) or can be derived from market prices of inflation swaps with different expiry dates.

Building Euro HICPx curve

As an example, we build an inflation curve for the euro-area for maturities up to 30 years as of trade date April 27, 2015. Euro zero coupon inflation swap rates are exchanged against change in Eurozone HICP excluding tobacco unrevised index subject to a 3-month lag (payoff at maturity of such swap is displayed in Figure 18). Unrevised numbers are the ones initially released by Eurostat, see Eurostat (2015). Published HICP indices are not seasonally adjusted. For illustration purposes, we extract monthly seasonal adjustment from the ECB calculations of seasonally adjusted HICPs ex tobacco[26] for the last 12 months (April 2014 – March 2015) using (6.13) (for ECB seasonal adjustment methodology we refer to ECB (2000)). We can see that in January and February prices are lower than inflation trend, while from March to June prices appear higher (Table 7). For simplicity, we assume that this monthly seasonality remains constant over time.

[24] If the seasonal component is annual then it needs to be divided by 12.

[25] X-12 ARIMA and newer X-13 ARIMA-SEATS is software for seasonal adjustment produced by US Census Bureau: https://www.census.gov/srd/www/x13as/

[26] The ECB produces seasonally adjusted HICP ex tobacco: http://sdw.ecb.europa.eu/browseSelection.do?ADJUSTMENT=S&node=2120778&DATASET=0&SERIES_KEY=122.ICP.M.U2.S.X02200.3.INX

In our example, the base index month is January 2015 (3 months lag) and index value equals 115.13[27]. Using quoted swap rates, we calculate projected index values from (6.10) and annual trend from (6.11), see Table 6.

Table 8. Projected CPI values and forward inflation unadjusted for seasonality.
Source: Bloomberg, Author's calculations

Tenor	Quoted breakeven rate	Maturity month CPI Index	Maturity date	Projected CPI	Annual trend
1Y	1.08%	Jan-16	27/04/2016	116.38	1.08%
2Y	1.00%	Jan-17	27/04/2017	117.44	0.91%
3Y	1.05%	Jan-18	27/04/2018	118.79	1.14%
4Y	1.13%	Jan-19	27/04/2019	120.43	1.37%
5Y	1.20%	Jan-20	27/04/2020	122.22	1.48%
6Y	1.26%	Jan-21	27/04/2021	124.10	1.52%
7Y	1.31%	Jan-22	27/04/2022	126.09	1.60%
8Y	1.35%	Jan-23	27/04/2023	128.20	1.65%
9Y	1.40%	Jan-24	27/04/2024	130.45	1.74%
10Y	1.45%	Jan-25	27/04/2025	132.89	1.85%
12Y	1.52%	Jan-27	27/04/2027	137.98	1.88%
15Y	1.61%	Jan-30	27/04/2030	146.25	1.94%
20Y	1.68%	Jan-35	27/04/2035	160.66	1.88%
25Y	1.74%	Jan-40	27/04/2040	177.21	1.96%
30Y	1.80%	Jan-45	27/04/2045	196.50	2.07%

Projected CPI curve is then used to calculate zero coupon inflation swap curve (Table 8 and Figure 19).

[27] Taken from Eurostat website:
http://ec.europa.eu/eurostat/tgm/table.do?tab=table&init=1&language=en&pcode=teicp240&plugin=1

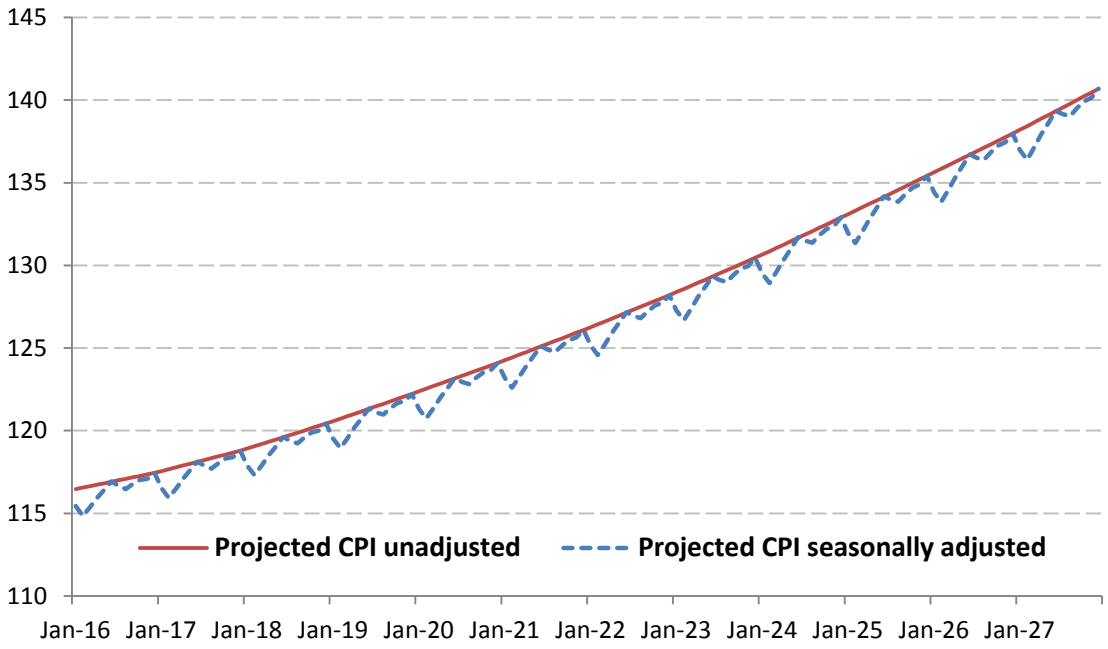


Figure 19. Unadjusted and seasonally adjusted CPI projections using (6.10) and (6.11).
Source: Author's calculations

Table 9. Calculation of monthly CPI projections adjusted for monthly seasonality and respective unadjusted and adjusted CPI curve (breakeven inflation rates).
Source: Author's calculations

Projected month	Maturity Date	Projected CPI unadjusted	Monthly seasonality	Projected CPI seasonally adjusted	Unadjusted breakeven inflation rates	Adjusted breakeven inflation rates
Jan-16	27/04/2016	116.4654	-0.88%	115.4450		
Feb-16	27/05/2016	116.5541	-0.60%	114.8418	1.14%	-0.23%
Mar-16	27/06/2016	116.6428	0.38%	115.3667	1.12%	0.18%
Apr-16	27/07/2016	116.7315	0.42%	115.9404	1.11%	0.56%
May-16	27/08/2016	116.8204	0.34%	116.4238	1.10%	0.84%
Jun-16	27/09/2016	116.9093	0.36%	116.9327	1.09%	1.10%
Jul-16	27/10/2016	116.9982	-0.31%	116.6594	1.08%	0.88%
Aug-16	27/11/2016	117.0873	-0.25%	116.4567	1.07%	0.72%
Sep-16	27/12/2016	117.1764	0.20%	116.7787	1.06%	0.85%
Oct-16	27/01/2017	117.2656	0.13%	117.0196	1.05%	0.93%
Nov-16	27/02/2017	117.3548	-0.02%	117.0852	1.05%	0.92%
Dec-16	27/03/2017	117.4441	0.23%	117.4441	1.04%	1.04%

In the following figure, a zero-coupon swap curve is constructed by calculating breakeven rates from projected CPI values.

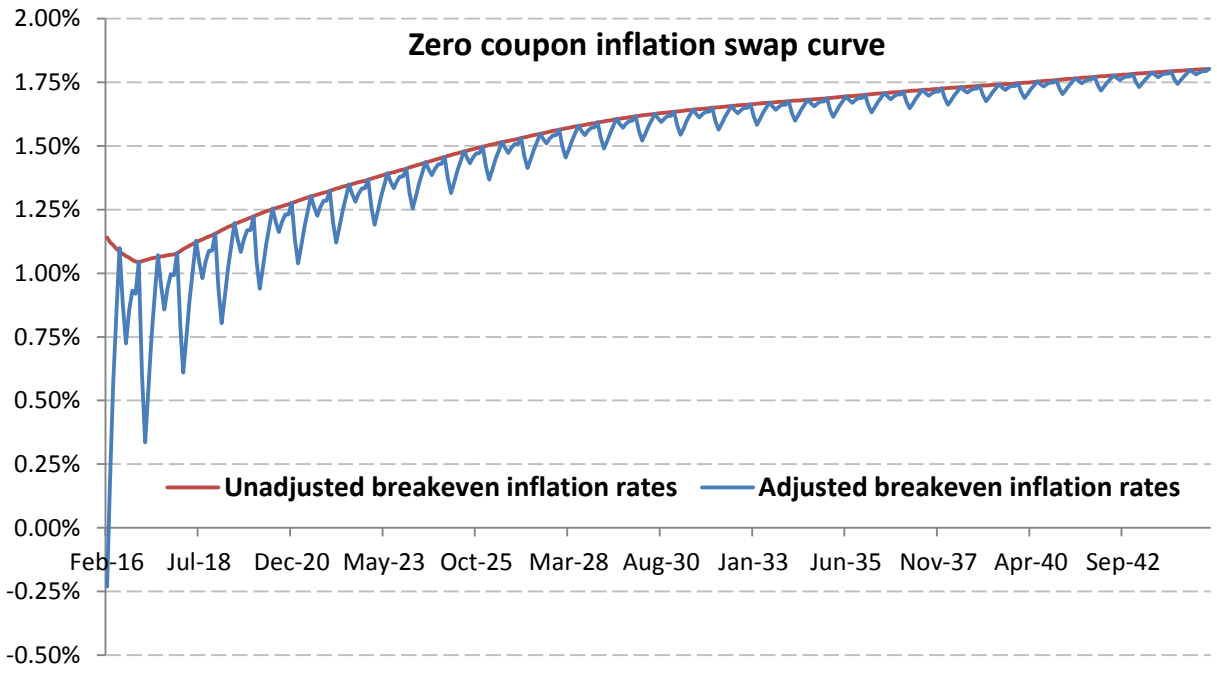


Figure 20. Interpolated zero coupon inflation swap curve, both unadjusted and adjusted for seasonality. *Source: Author's calculations*

Seasonal oscillation decreases in time and for longer maturities becomes less important. The effect of seasonality becomes stronger as swap or bond moves closer to maturity.

To conclude, we presented a consistent valuation approach to zero-coupon inflation swaps starting with a brief overview on the inflation market, its main participants, and explaining standard measurement of inflation and the notion of breakeven inflation rate. Then we discussed the real yield of inflation-linked bond versus the nominal yield of the traditional bond. Finally, we looked into mechanics of zero-coupon inflation swap and presented a valuation approach that is properly reflecting the risk of its cash flows. This approach is then illustrated in the case of zero-coupon inflation swap collateralized with cash using OIS discounting. The construction of inflation curve from zero coupon inflation swaps is outlined and illustrated on a practical example taking into account seasonality adjustments of inflation.

Conclusion

Interbank rates are not risk-free and neither are IBOR swap rates. The market has moved from discounting on IBOR swap rates to discounting on OIS rates and reflecting all economic costs of derivatives on top of OIS rates. OIS rates are generally considered as a good proxy for risk-free rates and their use is further justified by the fact that the interest rate paid on collateral is linked to OIS rate. Basis spreads in basis swaps or cross-currency basis swaps are here to stay; they impact the valuation and the discount factors and must be therefore part of the valuation equation. One curve can no longer be used to estimate forward rates and to discount future cash flows. In fact, changing the discount curve means changing the market value of a derivative. These are some of the general outcomes of the thesis. Others are more specific.

We discussed recently introduced valuation adjustments to derivatives that are needed to reflect both risk and costs of the transaction. Specifically, we showed how credit valuation adjustment (CVA), debit valuation adjustment (DVA), funding valuation adjustment (FVA), capital valuation adjustment (KVA), and initial margin valuation adjustment (MVA) step into the risk-neutral valuation of derivatives.

Then we focused on derivatives' discounting in detail. We derived discount factors and forward rates that are consistent with market prices and showed how basis swap spreads impact the valuation. We discussed in detail valuation of interest rate swaps, basis swaps and cross-currency swaps and showed why multiple curves need to be used to estimate forward rates and to discount future cash flows. In the last section we derived the approximation for discount curves implied from cross-currency market quotes and the discount curve of the base currency. We also decomposed basis spread into individual elements. We saw that CCS basis spread can be decomposed into basis spread that reflects credit and liquidity risk between the two currencies and an OIS basis spread reflecting demand and supply of one currency vs. another one. We used this decomposition to analyse basis spread, in particular, for the Czech derivatives' market in chapter 4.

We moved on to apply several of the findings on Czech derivatives market. We proposed an estimation of CZK OIS curve when not all OIS quotes are available. Interestingly, when we first investigated this topic, CZK OIS quotes have been only available up to one year and have only recently started to be quoted for maturities up to ten years. We assume there became a structural need for banks to pay and receive CZK OIS rates linked to CZK cash collateral earning CZEONIA rate. We also analysed EUR/CZK cross-currency basis swap spreads by decomposing them into individual components to find out that although the risk premium in CZK rates is higher than in EUR, the increased supply of CZK and higher demand for EUR via FX swap market

is pushing synthetic CZK rates into negative territory and it is the main factor that contributes to significantly negative EUR/CZK basis swap spreads. In the very long end (above 12 years) the main contribution to negative basis spread becomes the liquidity and credit risk premium of CZK rates. We also showed such breakdown can be used to approximate implied discount rates of the second leg of the cross-currency swap and may help optimize collateral posting for the counterparty that is out of the money.

We continued to analyse cross-currency basis spreads, this time, from a different angle. We discussed what is driving the credit and liquidity risk and supply and demand pressure of one currency versus another one. We built regression models for changes in 3-month and 5-year EUR/USD basis swap spread that explained 50% and 30% of the variance of weekly spread changes. The model confirmed, for example, that increased credit risk in European banks widens the EUR/USD basis spread. We then tested arbitrage-free boundaries for cross-currency funding and investing and identified several periods of arbitrage opportunities in cases of EUR/USD and EUR/CZK markets where for market participants able to raise unsecured funding in one currency and swapping it into another currency.

Finally, we focused inflation derivatives. We provided an overview of inflation markets and discussed inflation measurement, real yields, nominal yields, and the construction of inflation curve from zero coupon inflation swaps taking into account inflation seasonality. We also presented consistent valuation approach to zero-coupon inflation swaps reflecting the risk of its future cash-flows.

Understanding risk premiums in derivatives rates and pricing is helpful for decision making when acting in the market. The perception of risk and risk premium has changed over time. Although OIS rates have been generally accepted risk-free rate proxies and have replaced IBOR rates for derivatives' discounting (and they also reflect the interest rate paid on collateral), the new discounting poses a number of challenges. For example, we saw that the choice of collateral impacts market price, be it bond or cash, in one currency or another. OIS discounting is complicated because CSA annexes may be very different from one to another and often allow counterparties to post a number of different assets as collateral. They further have different threshold, margins, or minimum transfer amounts. This optionality impacts the pricing as well and banks generally opt to discount by the cheapest to deliver collateral. But this needs to be monitored because market factors change and so can the cheapest to deliver asset. More standardized CSA would certainly help[28] and regulators seem to be setting up rules to make

[28] In fact, ISDA published Standard CSA in June 2013 aimed at standardizing market practice in collateral management by removing optionality and promotes OIS (http://www2.isda.org/news/isda-publishes-2013-standard-credit-support-annex-scsa). Due to new regulatory changes (for example, Basel rules on margining as mentioned in Chapter 2), ISDA is now working on amendments to the Standard CSA.

central clearing of derivatives attractive also for those derivatives for which clearing will not be mandatory.

There also seems to be ambiguity in using and pricing different valuation adjustments. Some of them, e.g. MVA also raise computational challenges as their calculations may need a considerable amount of time. Although we have focused on valuation only, introducing valuation adjustment into practice is not only pricing it. It creates additional challenges for other departments such as risk management, reporting, or accounting. Discounting of derivatives has changed from a simple and straightforward task to sometimes impractically complex even for seemingly simple linear derivatives. Valuation of derivatives will certainly keep both academia and market community busy for the years to come.

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List of notations

Chapter 1

FRA	Forward rate agreement
IBOR	Interbank offered rate
IRS	Interest rate swap
OIS	Overnight Indexed Swap

Chapter 2

CVA	Credit valuation adjustment
DVA	Debit valuation adjustment
FVA	Funding valuation adjustment
KVA	Capital valuation adjustment
MVA	Initial margin valuation adjustment
R	Recovery rate
L	Discounted loss in case of counterparty's default
τ	Time of counterparty's default
ξ_τ	Counterparty exposure at time of default at time τ
DF_τ	Discount factor (present value of one unit payment that will be received at time τ)
$\mathrm{E}^{\mathrm{Q}}(\mathrm{DF}_t\xi_t\|\tau = t)$	Expected discounted exposure conditional on counterparty's default at time *t*
$\Pi_{0,t}$	Risk neutral probability of counterparty's default between time 0 and t
T	Final maturity date of the transaction
$\mathrm{E}\xi_{t_k}$	Expected exposure at time t_k.
$V_{i,t}$	Market value of derivative *i* at time *t*.
C_{t_k}	Value of collateral received at time t_k
ctpty	Notation of parameters linked to counterparty's default
own	Notation for one's own default.
s_t	Funding spread on top of one's own credit spread

Chapter 3

$DF_{d,t}$	Discount factor (present value of one unit payment that will be received at time *t*)
$DF_{OIS,t}$	Discount factor calculated from the OIS swap rates
$DF_{OIS_j,t}$	OIS Discount factor for the *j*-th currency in cross-currency basis swap*, j=1,2*
$DF_{j,t}$	Discount factor for the *j*-th currency in cross-currency basis swap*, j=1,2*

f_t	Implied forward reference rates of the floating leg of the swap at times *t-1* to *t*
$f_{OIS,t}$	Implied forward OIS rate of the floating leg of the OIS swap for period *t-1* to *t*
$f_{i,t}$	Implied forward rate for *i*-th leg of either basis swap or cross currency basis swap, *i*=1, 2
$\delta_{i,t}$	Year fraction for period between *t-1* and *t* for *i*-th leg (i=1,2)
D_t	Actual number of days between *t-1* and *t*
B	Number of days in a year according to the day count convention for the respective currency (e.g. for EUR, USD or CZK B=360)
$\mathrm{r_m}$	Quoted IBOR swap rate
$\mathrm{r_{ON}}$	Overnight (ON) money market cash rate
$\mathrm{r_{TN}}$	Tommorow next money market cash rate (overnight rate starting tomorrow)
$r_{OIS,n}$	Quoted OIS swap rate for maturity *n*
ONIA	OverNight Index Average (EONIA for EUR, Federal funds effective rate for USD and CZEONIA (Czech OverNight Index Average) for CZK).
ISDA	International Swaps and Derivatives Association
CSA	Credit support annex
CCP	Central counterparty
EMIR	European Market Infrastructure Regulation
bs_m	Basis spread for maturity *m* in basis swap
BS_m	Basis spread for maturity *m* in cross-currency basis swap
$BS_{OIS-OIS}$	Basis spread representing credit and liquidity risk between two currencies
BS_{FX}	IBOR-based cross-currency basis spread
$BS_{FX,OIS}$	OIS cross-currency basis spread

Chapter 4

$DF_{CZK\ OIS,t}$	Discount factor from CZK OIS rates
$r_{CZK\ FX,t}$	Swap rate implied from EUR/CZK basis swap spreads
S	FX spot rate
sp	Swap points
$BS_{CZK\ FX,OIS}$	Basis spread of EUR/CZK OIS cross-currency swap (EONIA vs CZEONIA + spread)
$BS_{EURCZK,OIS-OIS}$	Basis spread that shows credit and liquidity risk between CZK and EUR rates
$BS_{EURCZK\ FX}$	Basis spread of EUR/CZK cross currency basis swap

Chapter 5

β_i	Regression coefficients, *i=1,...,5*
ΔBS	Weekly change in cross-currency basis spread
$r_{FX,t}$	FX swap implied USD funding rate from EUR funding rate (Euribor)
$r_{FX\ EUR,2W}$	FX-swap implied EUR funding from CZK funding rate (Pribor)
X	Indicator of an arbitrage opportunity

Chapter 6

I_T	Price of the consumer basket at time *T*
I_{T_0}	Price of the basket at time T_0.
CPI	Consumer price index
HICP	Harmonized Index of Consumer Prices
I/L	Inflation-linked
N_{T_0}	Inflation-linked bond notional in year T_0
C_{T_0}	Inflation-linked bond real coupon in year T_0
DF_T^{IL}	Discount factor to calculate nominal present value of inflation-linked bond
DF_T^N	Discount factor to calculate nominal present value of standard bond
DF_T^R	Discount factor to calculate real present value of inflation-linked bond
r_T^N	Nominal yield of inflation linked bond
r_T^R	Real yield of inflation linked bond
b_T	Quoted zero-coupon swap fixed rate with maturity *T* years
$PR\ Index_{T+m}$	Projected CPI for date *T* years and *m* months after the start date
sc_i	Monthly seasonal component for *i-th* month after start month